

PRINCESS Trial Protocol

A double-blind placebo controlled trial to evaluate the efficacy of probiotics (*Lactobacillus rhamnosus*, LGG and *Bifidobacterium animalis* subsp. *lactis*, BB-12) in reducing antibiotic administration for infection in care home residents.

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General Information This protocol describes the PRINCESS trial, and provides information about the procedures for entering participants into the trial. The protocol should not be used as a guide, or as an aide-memoire for the treatment/care of other patients/participants. Every care has been taken in drafting this protocol; however, corrections or amendments may be necessary. These will be circulated to the known Investigators in the trial, but centres entering participants for the first time are advised to contact the South East Wales Trials Unit (SEWTU) in Cardiff to confirm that they have the most up-to-date version of the protocol in their possession. Problems relating to the trial should be referred, in the first instance, to SEWTU.

Compliance This study will adhere to the conditions and principles outlined in the EU Directive 2001/20/EC, EU Directive 2005/28/EC and the ICH Harmonised Tripartite Guideline for Good Clinical Practice (CPMP/ICH/135/95). It will be conducted in compliance with the protocol, the Research Governance Framework for Health and Social Care (Welsh Assembly Government November 2001 and Department of Health 2nd July 2005), the Data Protection Act 1998, and other regulatory requirements as appropriate.

For those trial participants lacking capacity, the Mental Capacity Act (2005) will be adhered to.

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Please contact the Trial Manager for general queries and supply of trial documentation

Randomisations:**Randomisation**

To randomise a patient log onto the following website:
Xx (TBA)
(See section 15 for more details)

Clinical queries:**Clinical queries**

All clinical queries should be directed to the Trial Manager who will direct the query to the most appropriate clinical person.

Serious Adverse Events:**SAE reporting**

Where an adverse event meets one of the serious categories an SAE form should be completed by the responsible person and faxed to the PRINCESS Trial Manager (See sections 13 for more details).

Fax Number: 02031070875

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Glossary of abbreviations

AAD	Antibiotic Associated Diarrhoea
AE	Adverse Event
AMR	Antimicrobial Resistant/ce
CAAD	Cumulative Antibiotic Administration Days
CAPD	Cumulative Antibiotic Prescription Days
CI	Chief Investigator
CID	Common Infectious Diseases
CIS	Consultee Information Sheet
CHR	Care Home Resident(s)
CRF	Case Report Form
CRS	Clinician Rating Scale
CTIMP	Clinical Trial of and Investigational Medicinal Product
CU	Cardiff University
EME	Efficacy and Mechanisms Evaluation Programme
EQ5D	EuroQol 5D
FBC	Full Blood Count
GCP	Good Clinical Practice
GI	Gastrointestinal Infections
GMP	Good Manufacturing Practice
GP	General Practitioner
HCAI	Health care associated infections
HI	Haemagglutinin Inhibition
IC	Informed Consent
ICECAP-O	ICEpop CAPability measure for Older people
ICF	Informed Consent Form
ICH	International Conference of Harmonisation
IDMC	Independent Data Monitoring Committee
MAR	Medication Administration Record
NHS	National Health Service
PAAD	Probiotics for Antibiotic Associated Diarrhoea study
PI	Principal Investigator
PIS	Participant/ Patient Information Sheet

PPI	Patient Public Involvement
QL (QoL)	Quality of Life
R&D	NHS Trust Research & Development Department
RCT	Randomised Controlled Trial
REC	Research Ethics Committee
RN	Research Nurse
RTI	Respiratory Tract Infection
SAE	Serious Adverse Event
SEWTU	South East Wales Trials Unit
TLR	Toll Like Receptors
TMG	Trial Management Group
TSC	Trial Steering Committee
UTI	Urinary Tract Infection
VDR	Vitamin-D Receptors
VRE	Vancomycin resistant enterococci

1 Amendment History

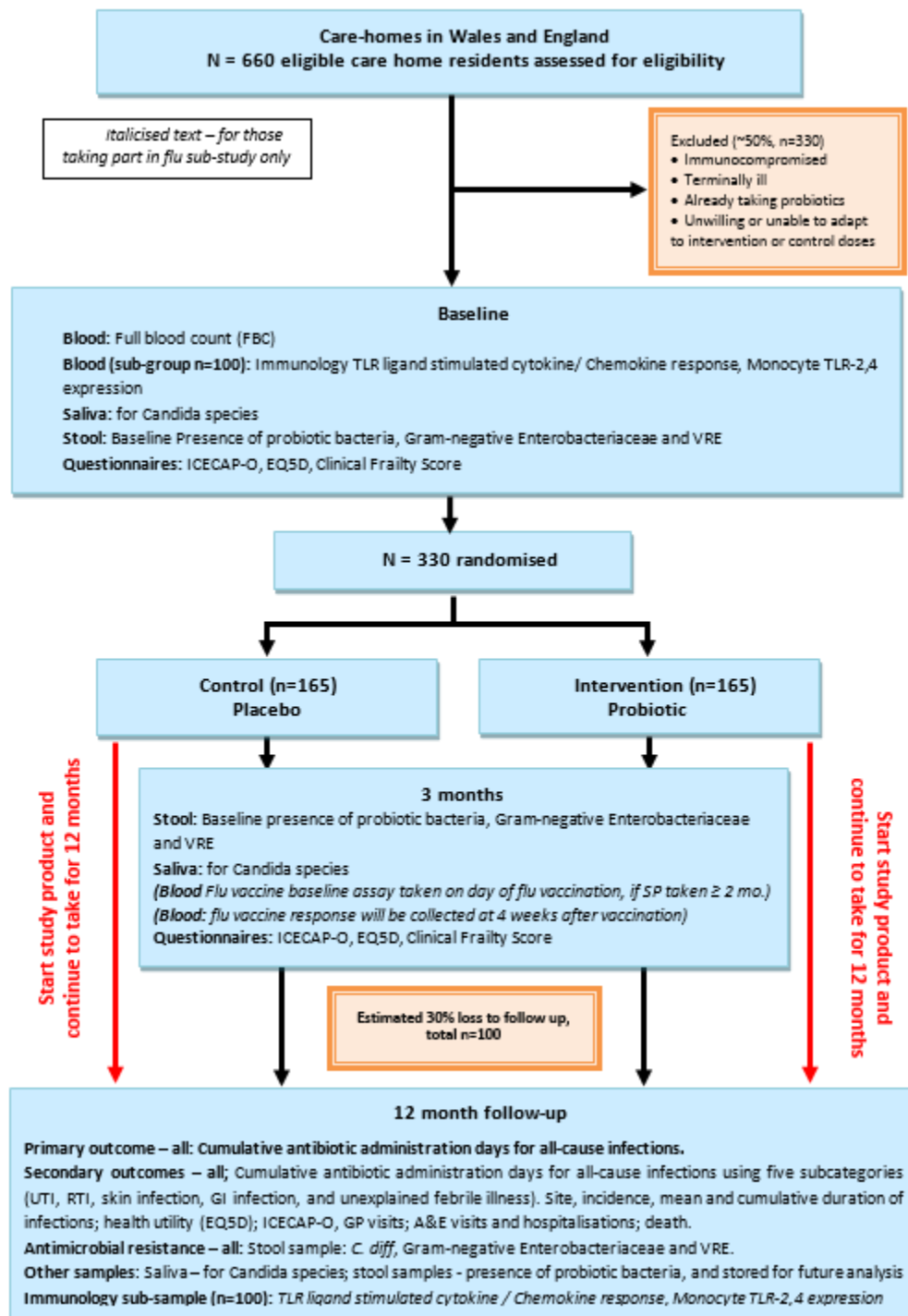
Amendment No.	Protocol version no.	Date issued	Author(s) of changes	Details of changes made

2 Synopsis

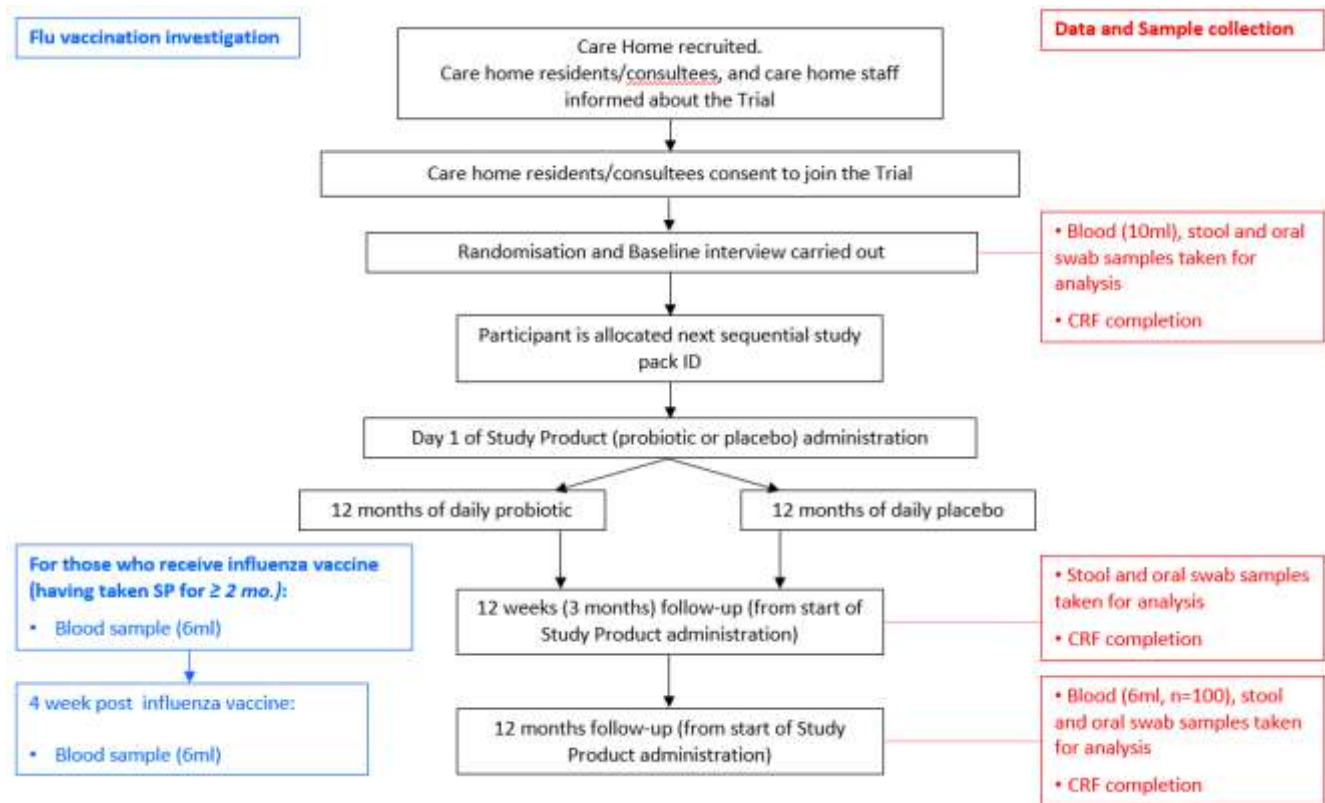
Short title	<u>Pro</u> biotics to <u>R</u> educe <u>I</u> nfections <u>i</u> n <u>C</u> are <u>E</u> home re <u>S</u> ident <u>S</u>
Acronym	PRINCESS
Internal ref. no.	
Trial/study design	Two arm, individually randomised, double blind placebo controlled efficacy trial.
Trial/study participants	Care home residents in Wales and England
Planned sample size	330
Follow-up duration	There will be two follow-up points – one at 3 months, and the last at 12 months
Planned trial/study period	Once daily oral probiotics or placebo for 12 months
Primary objective	To evaluate the effect of a dose of daily oral probiotics on cumulative antibiotic administration days (CAAD) for all-cause infections
Secondary objectives	There are secondary objectives in infection, health utility, wellbeing, service use, mortality, prophylactic antibiotic use, antimicrobial resistance, oral and gut microbiology, immunology, blood and mechanisms (see section 5).
Primary endpoint	CAAD for all-cause infections over 12 months
Secondary endpoints	CAAD for five sub-categories of CID (RTI, UTI, skin, GI unexplained fever); number, site, duration (mean and cumulative) of infection; EQ-5D (health utility), ICECAP-O (wellbeing); GP visits, A&E visits and hospitalisations; death; Antimicrobial resistance (AMR) (Culture and antibiotic sensitivity of Gram-negative Enterobacteriaceae and VRE from stool sample and presence of probiotic species in stool to explore adherence and contamination); Antibiotic consumption Mechanistic measures: Blood – FBC; Vitamin D; influenza vaccine response - Haemagglutination inhibition assay; immunology - TLR ligand stimulated (TLR2 and TLR4) cytokines/chemokines, receptor expression (circulating immune cells TLR2 and TLR4) Saliva – presence of oral Candida species Stool – presence of probiotic bacteria <i>Lactobacillus rhamnosus</i>, LGG and <i>Bifidobacterium animalis</i> subsp. <i>lactis</i>, BB-12, plus presence of AMR bacteria and <i>C. difficile</i> Tertiary outcome: To determine if level of vitamin D correlate with colonisation of AMR bacteria in faecal isolates (see section 5).
Interventions	Daily dose for 12 months of study product - probiotic or matching placebo

3 Trial summary and schema

3.1 Trial schema



3.2 Participant flow diagram



3.3 Trial summary

Care home residents (CHR) are prescribed far more antibiotics than the general population because of the high burden of infections they bear, caused by weakened immunity, close-proximity living and multi morbidity (1). In our previous research in care homes, we found CHR took antibiotics for an average of 17.4 days per year (2). High antibiotic use increases the risk of reservoirs of antimicrobial resistance (AMR) in care homes that can spread within care homes and to hospitals and the community (1). This will intensify as the population ages. Infections in CHR cost the NHS >£54 M/year in hospitalisation alone (3) and infections are the commonest reason for CHR to be hospitalised. AMR infections are generally more serious and costly, particularly in older people. Reduction in antibiotic use and AMR could improve quality of life, save money, and help preserve the usefulness of existing antibiotics.

Other than vaccination and hygiene methods, there are few interventions proven to prevent infection in CHR. Probiotics are live bacteria that may confer health benefit by improving immune function and reducing carriage of potentially harmful bacteria. Probiotics are safe and cheap and are available as supplements. Probiotics may mediate positive changes in gut bacteria and therefore decrease immunity decline and infection. Reviews of studies found that probiotics reduced antibiotic use and risk of respiratory tract infections in adults (4) and reduced antibiotic associated diarrhoea (AAD) (5). Probiotics also reduced carriage of AMR bacteria in a review of critically ill patients, and have reduced duration of CID (6) and enhanced immune response, including to flu vaccination (7). However, research in CHR is currently lacking.

The PRINCESS trial will trial a probiotic preparation containing *Lactobacillus rhamnosus*, LGG and *Bifidobacterium animalis subsp. lactis*, BB-12 to determine whether this product prevents infections over a 12 month period. Other probiotic species have not shown benefit in respiratory tract infection (RTI) or immune function in older people (8). We will assess total days on antibiotics for infections (including RTI, urinary tract infection (UTI), skin and gastro-intestinal infection (GI)), immune parameters (including influenza vaccine response), and changes in AMR in bowel flora. We will explore mechanisms, and test if probiotics reduce cumulative antibiotic prescribing days by 10% or more. To detect this we will recruit 330 CHR from care homes in Wales and England into an individually randomised trial of probiotic vs placebo, and follow them up for 12 months. We will seek informed consent from CHR with capacity, and consult with next of kin/representative for those lacking capacity.

4 Introduction

4.1 Background

Introduction

The PRINCESS trial will evaluate the efficacy and mechanisms of a nutritional intervention (daily probiotics) on cumulative antibiotic administration days (CAAD) for all-cause infections and reduce infection in care home residents. Antibiotic prescribing has been found to be associated with isolation of resistant organisms and subsequent infections that are resistant to antibiotics. Fewer and less severe infections may reduce the need for antibiotics in CHR. This may therefore reduce the driving influence on AMR and may reduce the likelihood of subsequent resistant infection.

Care home residents (CHR) are particularly vulnerable to common infectious diseases (CID) (1) which are the commonest cause of hospitalisation in this group, costing the NHS >£54 million/year, and often results in lasting health decline. Frailty, atypical presentation and bacterial colonisation patterns lead to high antibiotic use. Reducing antibiotic prescribing in the expanding care home sector is central to the challenge of containing antimicrobial resistance (AMR) in UK.

Probiotics

Probiotics are “live” microorganisms that, when administered in adequate amounts, confer a health benefit on the host. They are present in certain products available in supermarkets as foodstuffs and in formulations used for specific therapeutic purposes. They may prevent infection by blocking pathogenic colonisation and enhancing gut-immune interaction, with influence on mucosal and systemic immunity, leading to enhanced natural killer (NK) cell activity and vaccine response in the elderly (7). Probiotics are safe and well tolerated (9, 10), and systematic reviews found no serious adverse effects in participants of trials of probiotics for antibiotic associated diarrhoea. Some older people already use them regularly, despite an inadequate evidence base supporting their effect on CIDs. Probiotics are feasible to administer to CHR in the course of routine care: Carlsson and colleagues confirmed feasibility of serving a probiotic intervention to people in care homes with dementia for six months (11). They found it was easy to serve alongside usual diet, there were few side effects and that staff were able to complete the processes and measures. In contrast to antibiotic use, long term probiotic use does not result in resistance in commensal gut organisms (12).

The gastrointestinal tract may be a major reservoir for AMR bacteria. This is important because most urinary tract infections arise from auto-inoculation with organisms for the gut. We will directly assess gut colonisation with AMR bacteria, which may be reduced directly via probiotic-induced colonisation or indirectly via reduced cumulative antibiotic prescription days (CAPD). We will explore the underlying immunological mechanisms, including influenza vaccine response (diminished generally in CHR), probiotic modulation of toll-like receptor (TLR) expression and responses to *ex-vivo* pathogenic challenge.

Existing trials and the research gap

Small trials found benefits from probiotics on AMR colonisation, e.g. a systematic review of multidrug resistant gram negative organisms in critically ill patients (13) and a randomised controlled trial (RCT) of vancomycin resistant enterococci (VRE) in renal dialysis patients (14). A Cochrane review of

probiotics to prevent acute upper respiratory tract infections (URTIs) included 14 RCTs, ten of which were meta analysed (including a total of 3451 participants) (4). Probiotics reduced episodes of acute URTI (OR 0.58 95%CI 0.36-0.92) and antibiotic prescribing (OR 0.67, 95% CI 0.45-0.98). Side effects of probiotics were minor. The review noted poor allocation concealment of several studies, and heterogeneity, and recommended that future RCTs should “focus on older people.” This is because only four studies have compared probiotics to placebo in older people to prevent infections. Turchet and colleagues performed a single centre pilot study of otherwise well ‘free living’ elderly, randomised to receive a probiotic containing *L. casei* or placebo for three weeks (15). They found no differences in frequency of URTI (in this short study), but reduction in the severity and duration of URTIs. Guillemard randomised otherwise well people living in care homes in France to a probiotic containing *L. casei* or placebo for three months (6). Probiotics decreased the duration of CIDs, especially for URTIs. Makino and colleagues in Japan compared whether the intake of yogurt fermented with *Lactobacillus delbrueckii* over 12 weeks had an effect on the common cold, and found the risk was 2.6 times lower (OR 0.39) and there was increased in natural killer cell activity in the yoghurt group (16). However, randomisation and concealment may not have been adequate in this study. Van Puyenbroeck and colleagues found no difference in the duration of respiratory symptoms or the probability of respiratory symptoms in ‘healthy older people’ in nursing homes in Antwerp from a probiotic containing *L. casei* shirota for 176 days (8). However, relevant medical conditions and cognitive deficits were excluded. Antibiotic prescription rates were not reported. There was a high level of missing self-complete diary data that generated analytic challenges. The probiotic we plan to use will be different to the strain used in the Guillemard study. Effects of probiotics are thought to vary by strain due to differing resistance to gastric acid and bile, ability to colonise mucosa and susceptibility to antibiotics. Thus, overall, the evidence base supporting a recommendation for frailer elderly either to consume or not consume probiotics is insufficient to robustly guide care. Therefore, a new trial is warranted that is properly designed in terms of allocation and concealment, that focuses on antibiotic use of all cause infections in all CHR including those without capacity and who are most frail, with reliable, frequent external ascertainment.

PRINCESS will be the first rigorous efficacy RCT of daily probiotic vs. placebo probiotic over 12 months on cumulative antibiotic administration days (CAAD) for infections, and will provide data on cheap, safe and widely accessible interventions for the prevention of infection, antibiotic use and AMR in CHR. Whether or not the trial finds a positive effect, the results will help CHR make evidence based decisions either to take or not take this probiotic product in order to maintain their optimal health and wellbeing. Mechanisms will be explored, in terms of vaccine efficacy, cytokine/ chemokine response to TLR ligand stimulation and probiotic effect on cellular TLR expression. This will extend current mechanisms knowledge regarding the interplay between probiotics, gut microbiota and immune function.

Risks and benefits

Infections are a common cause of suffering and increased resource use in CRHs. Even so-called ‘minor infections’ such as urinary tract infections and upper respiratory tract infections can have an important negative impact on the health, wellbeing and dignity of older frail people. As people get older, the microbiological diversity in their gut reduces as they are less mobile, eat differently, and their immune systems become less active. Most urinary tract infections arise from infection by organisms in our own gut. By increasing the diversity of microbiological organisms in the gut, taking probiotics regularly as

a food supplement may reduce the prevalence of pathogenic bacteria and stimulate the immune system, adding to overall resilience and general wellbeing as well as reducing infections and antibiotic use. Health care associated infections (HCAI) include CIDs and cause significant debility, hospital admissions and death in CHR, burdening both the health service and care home staff. The risk of deterioration in general physical condition was twice as high in CHR in Norway who develop HCAI and were nine times more likely to be admitted to hospital (17). Ageing is independently associated with reduced immune response to infections (18). Probiotics reduced the incidence of RTIs in the general population, possibly by boosting immune mechanisms (4). Although probiotics carry theoretical risks of causing infection beyond the gut and transferring of antibiotic resistant genes, there have been no reports of bacteraemia or fungaemia attributable to the probiotics in trials (9, 10, 19). Gastrointestinal side effects and rash are generally no more common than in patients on placebo probiotic (5).

4.2 Rationale for current trial

The 2010 Adult Social Care Statistics reported that there were 229,900 people in residential care in England, with numbers predicted to steadily increase. The 2011 census reported that there were 291,000 people over the age of 65 resident in the care home population in England and Wales (20). CHR are particularly prone to infections. The year-long observational study (n=274) identified 609 infections that led to an antibiotic prescription (95% CI 1.90-2.46). The most common indications were RTI (47% of prescriptions), urinary tract infections UTI (29%) and skin infections (18%) (2). CID in CHR led to suffering, loss of dignity, hospitalisation, GP visits and death. Health needs assessment of 240 CHR in South Tyneside PCT identified 167 hospital admissions accounting for 1595 bed days over a year (21). On average, at least 4 beds in the Acute Trust are used for CHR all year round costing around £400K. About 25% of admissions were due to infection costing £100K in one PCT, or an estimated £54M/year for the UK. CIDs in CHR increase GP and care home burden and costs, impacting on opportunities for other aspects of care. Antibiotic prescribing leads to HCAIs such as *C. difficile* and drives AMR. The greatest risk of being infected with a AMR compared to a sensitive bacteria is recent consumption of antibiotics, even after controlling for age, comorbidity and other risk factors (22). AMR infections are more serious, last longer and are more costly to manage (23). Probiotics are effective in preventing AAD (5, 19), but we have no adequate data on prevention of all-cause common infections and antibiotic prescribing in care home residents.

There is an urgent need to reduce AMR through infection prevention in care homes. Urinary tract infections (UTIs) are usually caused by auto-inoculation; that is infection from our own bowel organisms. Carriage of AMR bowel organisms increases the chances of an AMR UTI. AMR Gram-negative septicaemia and AMR UTI are on the increase in the community, especially among older people (24). Care homes are a reservoir for AMR that cycle between the community and hospitals (3, 25). There is a steady increase in the care home population, and evidence-based interventions are needed to improve their quality of life through reducing the incidence of CIDs and antibiotic use (26).

In PRINCESS we will study the long-term effect of administering a combination of two probiotics *Lactobacillus rhamnosus* GG and *Bifidobacterium animalis* subsp. *lactis*, BB-12.

Lactobacillus rhamnosus GG has the most extensive number of human studies in a wide variety of populations including elderly individuals (27). These strains have been found to be safe. Although the effect of these probiotic strains on infections in care home residents or older people has not yet been

assessed, relevant evidence proving efficacy in other populations supports its use in this trial. A meta-analysis of RCTs involving 1805 children in total found that *Lactobacillus* GG was associated with reduced risk of otitis media, upper respiratory tract infections and antibiotic treatments (28). This meta-analysis included a double blind, placebo controlled RCT of 742 hospitalised children found *Lactobacillus rhamnosus* GG reduced the risk of gastrointestinal (RR 0.40; 0.25-0.70) and respiratory tract infections (RR 0.38; 0.18-0.85) and duration of these infections (29). An RCT of 281 children in day care found that *Lactobacillus* GG reduced the risk of respiratory tract infections over three months (RR 0.66; 0.52-8.82, NNT=5), and reduced the time with a RTI (30). A double blind RCT found that *Lactobacillus* GG acts as an immune adjuvant to influenza vaccination as measured by levels of protective antibodies to the H3N2 flu strain. This study stresses the need for future studies of probiotics as immune adjuvants focusing on groups known to have poor response to influenza vaccination (31). When administered orally as lozenges the combination of probiotics has been found to have beneficial effect on oral and dental health: reducing plaque and gingival inflammation without affecting oral microbiota (32).

Many studies include more than one probiotic strain in the intervention, and in several studies, *Lactobacillus* GG and *Bifidobacterium animalis* subsp. *lactis*, BB-12 have been included in the same probiotic formulation (33, 34). A randomised placebo controlled trial of 231 college students taking a combination probiotic including *Bifidobacterium animalis* subsp. *lactis*, BB-12 for 12 weeks found that duration and severity of upper respiratory tract infections were improved by the active probiotic (35). A randomised, placebo-controlled, double blind, parallel dose-response study investigated the impact of 4-week commercial yoghurt consumption supplemented with *Bifidobacterium animalis* subsp. *lactis* (BB-12). The probiotic strain remained active during gut transit and was associated with an increase in beneficial bacteria and a reduction in potentially pathogenic bacteria (36). In a double-blind, placebo- controlled study, 109 new-born 1-month-old infants receiving *Bifidobacterium animalis* subsp. *lactis* BB-12 were reported to have experienced fewer respiratory infections (65 v. 94 %; risk ratio 0.69; 95 % CI 0.53, 0.89; P = 0.014) than the control infants (37). A randomised, double-blind, placebo-controlled study Infant formula supplemented with the probiotics *Lactobacillus rhamnosus* GG and *Bifidobacterium lactis* Bb-12 or placebo was administered daily until the age of 12 months. Those receiving the active probiotic had fewer episodes of otitis media, received fewer courses of antibiotics in the course of routine care, and had fewer recurrent respiratory infections (38). A multispecies probiotic supplementation that included *Lactobacillus rhamnosus* GG and *Bifidobacterium animalis* ssp. *lactis* Bb12 reduced symptoms of irritable bowel syndrome and stabilised the bowel microbiota (39). A randomised, double-blind, placebo-controlled trial in 12 wards in two nursing homes in Finland involving 209 residents received either 109 CFU/day *Bifidobacterium longum* strains or 2) 109 CFU/day *Bifidobacterium lactis* Bb12 or 3) placebo for 7 months. Compliance was 85%, and the groups receiving active products had more frequent bowel movements than the placebo group (40). In a 7-week double-blind crossover study, 36 subjects were randomised to receiving yoghurt with *Bifidobacterium animalis* subsp. *lactis* BB-12 or placebo. BB-12 was safe. The defecation frequency during the BB-12 intake period was increased compared to the no-intake period for group, and comfort after defecation improved significantly (41).

Thus, while both of the proposed probiotic strains have never been evaluated for the purposes intended in the PRINCESS trial; there is evidence from other population groups that these strains have the effect of reducing frequency, severity and duration of infections and stabilising bowel flora. Many

infections in the elderly result in autoinoculation from the bowel. It could be likely that the two strains may supplement each other in terms of efficacy due to different mode of actions.

5 Trial objectives and outcomes

HYPOTHESIS (Primary): Daily oral probiotic reduces CAAD for infection vs placebo in CHR.

HYPOTHESIS (Mechanisms): Daily oral probiotic reduces gastrointestinal colonisation with AMR bacteria, enhances influenza vaccine response and modulates ex-vivo cytokine response to Toll Like Receptor (TLR) agonists vs placebo in CHR.

5.1 Primary and secondary objectives and outcome measures

	Objectives	Outcome Measures	Timepoint(s) of evaluation of this outcome measure
Primary	To evaluate the effect of a daily dose of oral probiotics on cumulative antibiotic administration days (CAAD) for all-cause infections	CAAD for all cause infections; Total number of days of systemic antibiotic administration as recorded in care home medical records and hospital medical records if the participant is admitted to hospital	Over a 12 month period (data recorded at regular intervals by RN from care home records)
Secondary	To determine the effect of daily probiotic intake;		
Infection	On CAAD for the following sub-categories of infection; respiratory tract infections (RTI), urinary tract infections (UTI), skin infections, gastrointestinal infections (GI), unexplained fever and other	Total number of days of antibiotic administration for each infection type as recorded in care home medical records (collect generic name and mode of delivery) (to be recorded as defined daily dose multiples) Number, site, duration (mean and cumulative) of infection. Use of antimicrobials	Over a 12 month period (data recorded at regular intervals by RN from care home records)
	The site, incidence and duration of infection		Over a 12 month period (data recorded at regular intervals by RN from care home records)
	On prevalence of <i>C. difficile</i> infection (clinical and bacteriological evidence of <i>C. difficile</i> colonisation)	Stool sample laboratory analysis	At baseline, 3 and 12 months

Health Utility	On patient's health utility	Self and/or proxy reported health related quality of life measurement EQ5D (5L)	At baseline, 3 and 12 months
Wellbeing	On patients wellbeing	Self and/or proxy reported ICEpop CAPability measure for Older people (ICECAP-O)	At baseline, 3 and 12 months
Service Use	NHS service use	In relation to infections; number of unscheduled GP consultations triggered by an acute episode (exclude routine/review GP visits), number of A&E visits, number of unscheduled hospital visits and/or stays for all-cause hospitalisation (as recorded in care home records and discharge summaries)	In a 12 month period (data recorded at regular intervals by RN from care home records)
Mortality	Mortality rates	Total number of deaths of trial participants (collected from care home records)	In a 12 month period (data recorded at regular intervals by RN from care home records)
Prophylactic antibiotic use	To record use of prophylactic antibiotics	Total number of days of prophylactic antibiotics administration as recorded in care home medical records and hospital medical records if the participant was admitted to hospital	In a 12 month period (data recorded at regular intervals by RN from care home records)
Antimicrobial resistance	Gastrointestinal colonisation with antimicrobial resistant (AMR) bacteria	Culture and antibiotic sensitivity of Gram-negative Enterobacteriaceae and vancomycin resistant enterococci (VRE) from stool sample	Samples taken at baseline, 3 and 12 months
Oral microbiology	Levels of oral Candida	Semi quantitative analysis of oral rinse or saliva samples	Samples taken at baseline, 3 and 12 months
Influenza vaccine response (for those who have received influenza vaccine only)	Influenza vaccine efficacy	Blood sample - Haemagglutinin inhibition assay and antibody titers	Sample taken on the day of (or up to 10 days prior to), and 4 weeks (28 days) post influenza vaccination (trial participants need to have been on Study Product for at least 2 months prior to influenza vaccination to take part in this sub-study)
Immunology (ex-vivo responses to pathogenic challenge) (n≈100)	Patient's cytokine and chemokine response in whole blood samples stimulated ex-vivo by TLR2 and TLR4	Response measured by laboratory analysis in whole blood (stimulated ex-vivo by TLR2 and TLR4)	Baseline and 12 months

	Modified expression of Toll Like Receptors (TLR)	TLR receptor expression measured in circulating white blood cells	Baseline and 12 months
Gut microbiology	Microbiome composition analysis and phylotype	Investigative work to analyse level of <i>Lactobacillus rhamnosus</i> , LGG and <i>Bifidobacterium animalis</i> subsp. <i>lactis</i> , BB-12 from stool sample	Samples taken at baseline, 3 and 12 months
Haematology and Biochemistry	Haematology; biochemistry	Full blood count (FBC); vitamin D	FBC and Vitamin D at Baseline (all participants); FBC on n≈100 (immunology participants only) at 12 months
Tertiary	To determine if the level of vitamin D at baseline correlate with colonisation of AMR bacteria in faecal isolates	Level of calcium and AMR colonisation within faecal sample	Baseline
Additional baseline measurements		Demographic information (to include age); clinical frailty score; use of proton pump inhibitors, laxatives and dose of Vitamin D	Baseline
Other		Record increased level of care required	Over 12 months from MARS sheet
		Record whether trial participant refused to give blood/saliva/stool sample	Over 12 months from care home resident record
		Ask care home staff what arm they think the trial participant is in	12 month CRF
		Record most common method of delivery of study product e.g. swallowed with water or sprinkled on food/drink	Over 12 months from MARS sheet

All tissue samples donated by trial participants will be taken on the understanding that they may be used for future research projects in the UK. The samples will be stored and subsequently distributed to approved projects in accordance with the Human Tissue Act and ethical legislation. This statement will form part of the consent procedure.

6 Trial design

A double blind, individually randomised two arm trial design to assess the effect of a daily oral probiotic versus placebo on CAAD for infection in 330 Care Home Residents.

6.1 Internal Pilot

A three month internal pilot phase will assess:

- CHR recruitment rate assumptions
- The suitability of the data collection tools and processes
- Study product administration rate
- The robustness of the sample collection and shipment processes

Information gathered during this period will be used to inform any changes to the main trial design or procedures. It is not anticipated that any substantial changes will be required. All trial procedures, participant visit and sample collection detailed below will be carried out during the internal pilot. All participants will be followed up for as long as possible to a maximum of 12 months.

We anticipate recruiting four care homes for the internal pilot – approximately two in the South Wales area and approximately two in the Oxford area. These four care homes should have at least 30 residents in each home.

7 Care Home and Investigator selection

We anticipate that a total of approximately 20 care homes located in Wales and England will be recruited to the trial. We will aim to recruit larger care homes with a minimum of 50 CHR in order to organise and plan recruitment. Care homes with less than 50 CHR may be selected if they have adequate recruitment potential. The manager of the care homes will be approached to obtain permission for the care home to take part in the trial and for all CHR to be approached about the trial.

The senior care home staff / nursing staff will ensure that the following documents have been received by the coordinating centre (see contact details on page 2):

- Site Specific Assessment approval for the care home
- A signed trial agreement (care home lead and sponsor signature)
- Completed signature list and roles and responsibilities document
- Completed contacts list of all site personnel working on the trial
- Participant/consultee information sheets and consent/declaration forms

Upon receipt of all the above documents, the coordinating centre will send a confirmation letter to the care home lead detailing that the care home is now ready to recruit participants into the trial. The coordinating centre will also provide the care home with a site file, in which the confirmation letter should be filed. The care home will also be provided with all documents required to recruit a participant into the PRINCESS trial and trial supplies.

8 Participant selection

8.1 Trial population

Care home residents in Wales and England.

Participants are eligible for the trial if they meet all of the following inclusion criteria and none of the exclusion criteria. All queries about patient eligibility should be directed to the coordinating centre before randomisation.

8.2 Inclusion criteria

- Currently living in a care home setting (residential, nursing or mixed)
- Participant is willing and able to give informed consent for participation in the trial OR if the participant lacks capacity, a consultee is willing to complete a consultee declaration form
- Aged 65 years or older

8.3 Exclusion criteria

The CHR may not enter the trial if **ANY** of the following apply:

- Is known to be immunocompromised (requiring immunosuppressants, long term high dose oral, intramuscular or intravenous steroids)
- Is currently taking regular probiotics and is not willing to adapt to trial protocol
- Is a temporary care home resident (i.e. less than 1 month of planned transitional/respite residential care)
- Death is thought to be imminent
- Lactose intolerant

9 Outcome measures

9.1 Primary outcome measure

Please see section 5 Trial objectives and outcomes.

9.2 Secondary outcome measure

Please see section 5 Trial objectives and outcomes.

10 Recruitment and randomisation

10.1 Number of participants

A total of 330 participants will be recruited.

10.2 Recruitment process

The PRINCESS trial will randomise 330 care home residents (CHR) to either receive probiotics or placebo treatment for 12 months. In order to achieve this target we estimate that 660 care home resident from approximately 20 care homes will need to be invited to participate in the trial. Care home staff, nursing staff and/or PRINCESS Trial Research Nurses (RNs) will identify those CHR who are potentially eligible to join the trial.

A screening log of all ineligible and eligible but not consented/not approached CHR will be kept at each care home so that any biases from differential recruitment can be explored. The screening log should be sent to the coordinating centre on request (see section 20 for further detail on data monitoring/quality assurance).

The participant's GP will be informed of the participant's entry into the trial. Each GP will be provided with a summary of the trial and the participant information sheet (PIS) plus any other information they may require as appropriate. It will be made clear that the research team will collect data on infections but that all clinical assessments and management of any infections will remain the responsibility of the GP.

Care homes will be provided with information on prevalence of *C. difficile* infection arising at their care home.

Each Care Home will receive £750 to cover their involvement – this will consist of £400 to help with set-up costs, and £350 six months later if participants have been recruited.

10.3 Informed consent

There are two categories of CHR who are eligible to join the trial; those who have capacity and are able to consent for themselves and those unable to consent for themselves (lack capacity). Consent procedures will differ according to the mental capacity of the CHR. Informed consent or a consultee declaration (see below) must be obtained prior to any trial procedures being undertaken.

The following information sheets apply:

Participant

- Pictorial Participant Information Booklet
- Participant Information Booklet
- Participant Consent Form

Consultee

- Consultee Information Booklet

- Consultee Declaration Form

Mental Capacity Assessment: Mental capacity will be assessed according to the Mental Capacity Act 2005, which provides a legal framework within which health and care professionals must act. As such all care homes have senior staff members who are fully trained to assess the mental capacity of CHR at any time. This Code of Practice (42) provides comprehensive advice on good practice for the assessment of capacity, which is time and decision-specific and depends on clinical judgement within a valid contestable process.

According to the Mental Capacity Act, the CHR will be assumed to have capacity unless it is established that they lack capacity, all practicable steps having been made to help them do so. Where there is a concern that a CHR lacks capacity to provide informed consent for themselves to participate in the trial, the CHR will be assessed for mental capacity by delegated individuals (e.g. senior care home staff/nursing staff/RNs). A standard template for recording of the mental capacity assessment will be provided to care homes, together with specific training on the use of this template. This document will be reviewed by a delegated individual (e.g. senior care home staff/RN) to enable a decision to be made regarding the mental capacity of the CHR, prior to taking consent from the CHR or a consultee declaration from their representative if the CHR is assessed as lacking capacity.

For both CHR with capacity and those lacking capacity: All potentially eligible CHR or their consultee will be fully informed about the trial through the PIS or CIS respectively supplemented with verbal explanations. The detailed PIS and CIS will include: the exact nature of the trial; what it will involve for the participant (and consultee, if applicable); the implications and constraints of the protocol; the known side effects and any risks involved in taking part. It will be clearly stated that the participant is free to withdraw/be withdrawn from the trial at any time for any reason without prejudice to future care, and with no obligation to give the reason for withdrawal.

The participant or consultee will be allowed as much time as they need to consider the information, and the opportunity to ask questions. Written Informed Consent (or exceptions as listed below) or Consultee Declaration will then be obtained by means of participant or consultee dated signature, respectively, and dated signature of the person who presented and obtained the Informed Consent/Consultee Declaration. The consent/declaration process will be undertaken by a suitably qualified and experienced delegated individual who has been authorised to do so by the Chief/Principal Investigator (e.g. care home staff/nursing staff/RN). A copy of the signed Informed Consent or Declaration Form will be given to the participant or consultee, respectively. The original signed form will be retained at the trial site.

Eligible CHR may participate in the trial even if they (or their consultee) prefer to opt out of the participant providing blood and/or stool and/or saliva samples.

Specific consent will be sought at trial entry from CHR with capacity to remain in the trial should capacity be lost during the trial.

CHR lacking capacity: Where a CHR lacks capacity to provide consent for themselves, a legal representative/guardian will be consulted who acts as their 'consultee'. The consultee may be a friend

or unpaid carer or relative, an attorney acting under a lasting power of attorney, or a court appointed deputy who has a relationship or personal knowledge of the person who lacks capacity. The consultee will be provided with information about the trial and they will be asked their advice regarding the inclusion of the person and what the person's views would have been regarding inclusion in the research if they had capacity to make the decision for themselves. In the event that a consultee who has an unpaid or non-professional role in caring for the person cannot be identified, or is not willing to act, a nominated consultee will be appointed and consulted prior to including the resident in the trial.

In the event that a consultee themselves loses capacity or dies during the trial period, an alternative representative will be contacted to act as consultee and a new consultee Declaration Form will be completed. If there is no one to represent the CHR in the trial, the CHR will be excluded or withdrawn depending on whether a CHR has already been recruited to the trial.

If a participant regains capacity during the trial period, the participant will be fully informed about the trial and informed consent to remain in the trial will be obtained from the participant.

Consultee Declaration by post: If a consultee is required and they cannot attend a face-to-face interview for the declaration process, the above documents may be sent by post from the care homes. The delegated individual (e.g. care home senior staff/RN) will countersign the signed and dated consultee declaration form returned by the consultee and a copy of the completed consultee Declaration Form will then be sent to the consultee. The original consultee Declaration Form will be retained at the trial site. All consultees who have not responded will be contacted by phone at least once by the individual delegated to undertake the declaration process.

Verbal consent/declaration: In the event that a CHR with capacity or a consultee for a CHR lacking capacity cannot provide handwritten signatures on the Consent/Declaration Form, verbal consent will be taken. In such cases, a delegated individual (e.g. senior care home staff) will read and discuss the trial with the CHR or their consultee to ensure understanding of the trial protocol. A member of the research team will witness, sign and date the Consent/Declaration Form to approve that consent or a consultee declaration has been given.

10.4 Randomisation/registration and unblinding

Participants will be remotely randomised using an online computerised randomisation system created by the University of Oxford Primary Care Clinical trials Unit (PCCTU). The system will be operational 24 hours a day.

Randomisation will be performed by the RN only after the participant has signed the Consent Form (or their consultee has signed a consultee Declaration Form) and completed the baseline assessments.

The RNs will be provided with an individual login ID to the online system. Participants will be randomised to either probiotic or placebo.

As PRINCESS is a double-blind trial, the participants, care home staff, treating clinicians, and trial team (including the trial statistician and RNs conducting all assessments) will be unaware of the group to which the participant has been allocated for the duration of the trial.

Each Study Product pack (probiotic or placebo) will be labelled with a unique identification number (pack ID). The online system will allocate a pack ID for each participant. The pack will contain one month's supply of study product (see section 12).

The participant ID should be entered on the pack label and both the participant ID and pack ID should be entered onto the CRF by the randomiser. The coordinating centres will also be notified that a participant has been randomised via an automated e-mail alert mechanism.

When a new study product pack is required (i.e. on a monthly basis): the delegated individual should enter the participant ID into the online system and will be allocated another pack ID. The participant ID should then be entered onto the pack label and the pack ID entered onto the paper CRF.

The unique participant IDs and pack IDs will be linked in the randomisation file, which will only be accessible by a statistician who is independent of the trial.

For more details, please consult the PRINCESS randomisation protocol.

Randomisation

To randomise a patient log on to <<website URL>>

Unblinding: There will not be an emergency unblinding procedure – care homes will be advised to stop study product administration if this is necessary. Unblinding will be available during normal office working hours.

In the event that the participant needs to be unblinded, the care home staff are directed to contact the PRINCESS team on 02920 687601. The PRINCESS unblinding SOP will be followed to unblind the participant and in this case the unblinding of a randomised trial participant can only be carried out by the QA Manager at SEWTU.

10.5 Screening logs

A screening log of all ineligible and eligible but not consented/not approached will be kept at each centre so that any biases from differential recruitment will be detected. The screening log should be sent to the PRINCESS Trial Manager (see section 20 for further detail on data monitoring/quality assurance).

11 Withdrawal & loss to follow-up

Each participant/consultee has the right to withdraw from any aspect of the trial at any time. The participant's care will not be affected at any time by declining to participate or withdrawing from the trial. Clear distinction must be made as to what aspect of the trial the participant is withdrawing or is withdrawn from. These aspects could be:

- Withdrawal from entire trial and does not want any data or samples already collected relating to them to be used
- Withdrawal from trial, study product and all subsequent trial follow-up (sample collection and data collection), but data and samples already obtained up to this point can be used
- Withdrawal from study product but happy to continue with follow-up processes (to include data collection and/or sample collection) – this may be due to a request to withdraw, or as a result of a change in the participant's condition or circumstances which justifies the discontinuation of the study product
- Withdrawal from study product and all subsequent sample and questionnaire data collection, but happy for routine care record data to be collected.

If a participant wishes to withdraw from the trial or is withdrawn all trial data collected to that date will be included in the final analysis subject to the appropriate consent being in place. The reason for participant withdrawal will be detailed in a Case Report Form (CRF) and reviewed by the CI and independent committee. Any queries relating to potential withdrawal of a participant should be forwarded to the coordinating centre immediately.

12 Intervention

Participants in the PRINCESS trial will be asked to take an oral dose of probiotic (*Lactobacillus rhamnosus*, LGG and *Bifidobacterium animalis* subsp. *lactis*, BB-12) or a matched placebo once daily for 12 months. The probiotic or placebo (referred to as Study Product) will be administered by the CHR's normal care giver and provided in capsule form. The preferred route of administration will be in the following order:

- 1) The capsule swallowed whole with water
- 2) The capsule emptied into a small amount of cold or lukewarm liquid and then swallowed
- 3) The capsule opened and its contents sprinkled onto cold or lukewarm food (not hot food) and then eaten

Suitability for religious groups and dietary preferences

The capsules are suitable for vegans, are Halal and Kosher (Kosher dairy excluding Passover), do not contain any genetically modified organisms or genetically modified raw materials, and allergen labelling is not required. The capsules do contain lactose. Care homes will be provided with documentation detailing the composition of the capsules.

The capsules are stable at room temperature for two years and temperature monitoring is not required for short term (e.g. one month) storage.

Labelling and recording

The capsules will be provided to the care home labelled with a unique pack ID. Packs will be allocated to participants on a monthly basis. The delegated individual (e.g. RN) should enter the participant ID into the online pack allocation system (see section 10.4) and will be allocated a pack ID. The participant ID should then be entered onto the pack label and the CRF completed.

Participants admitted to hospital would not be expected to continue taking the study product during their hospital stay.

12.1 Adherence

Data regarding participant's adherence to the study product will be collected from a number of sources:

- MAR sheets - administration of the study product will be recorded by care home staff using Medication Administration Record sheets (MAR sheets) and monitored by RNs during their visits to each care home. MAR sheet are used in most care homes and will allow incorporation of trial procedures into routine practice easily. If MAR sheets or an equivalent recording system is not used routinely by any care home participating in the PRINCESS trial the research team will provide MAR sheets and training as appropriate.
- The Clinician Rating Scale (CRS) will also be used to record the level of adherence shown by a participant.
- Presence of probiotic organisms in bowel - the presence of the probiotic organisms will be assessed in stool samples of participants at baseline, three and 12 months. This will give an indication of adherence, survival of the probiotic in the large bowel and potential contamination in the placebo arm.
- Capsule count - regular counts of unused study product will be undertaken by the PRINCESS trial research nurses.

13 Adverse Events

Adverse Event (AE): Any untoward medical occurrence in a trial participant which does not necessarily have a causal relationship with this treatment. An AE can therefore be any unfavourable and unintended sign (including abnormal laboratory finding), symptom, or disease.

The probiotic being used in this study is a well-established food supplement made to GMP (Good Manufacturing Practice) standards with an unblemished safety record. We will therefore, not be collecting AEs.

Serious Adverse Event (SAE): Any adverse event that:

- Results in death
- Is life-threatening*
- Required hospitalisation or prolongation of existing hospitalisation**
- Results in persistent or significant disability or incapacity
- Consists of a congenital anomaly or birth defect
- Other medically important condition ***

* Note: The term “life-threatening” in the definition of serious refers to an event in which the participant was at risk of death at the time of the event; it does not refer to an event which hypothetically might have caused death if it were more severe.

** Note: Hospitalisation is defined as an inpatient admission (at least an overnight stay), regardless of the length of stay, even if the hospitalisation is a precautionary measure, for continued observation. Pre-planned hospitalisation e.g. for pre-existing conditions which have not worsened or elective procedures does not constitute an adverse event.

*** Note: other events that may not result in death are not life-threatening, or do not require hospitalisation may be considered as a serious adverse event when, based upon appropriate medical judgement, the event may jeopardise the participant and may require medical or surgical intervention to prevent one of the outcomes listed above.

13.1 Serious Adverse Events

This study is not a CTIMP and this population will have a vast number of health events in the normal course of their care at this stage in their lives. Since the probiotic being used is a well-established food supplement, we will only collect a well-circumscribed set of SAEs in this trial from first study product administration for the 12 month follow-up period.

For the purpose of this study, a **Serious Adverse Event (SAE)** will only be reported if it is classified as being probably or definitely related to specific effects of the study product such as bloating, septicaemia due to presence of probiotic bacteria, or ingredient-related, in which case it is a **Serious Adverse Reaction (SAR)**. SEWTU will notify the Sponsor of all SARs.

All other incidences of, and the reason for, hospital admission and death will be recorded and reported as part of routine follow-up using the follow-up CRF and reported (annually) as **unrelated Serious Adverse Event (SAEs)**.

13.2 Causality

The assignment of the causality should be made by the Chief Investigator or clinical reviewer using the definitions in the table below.

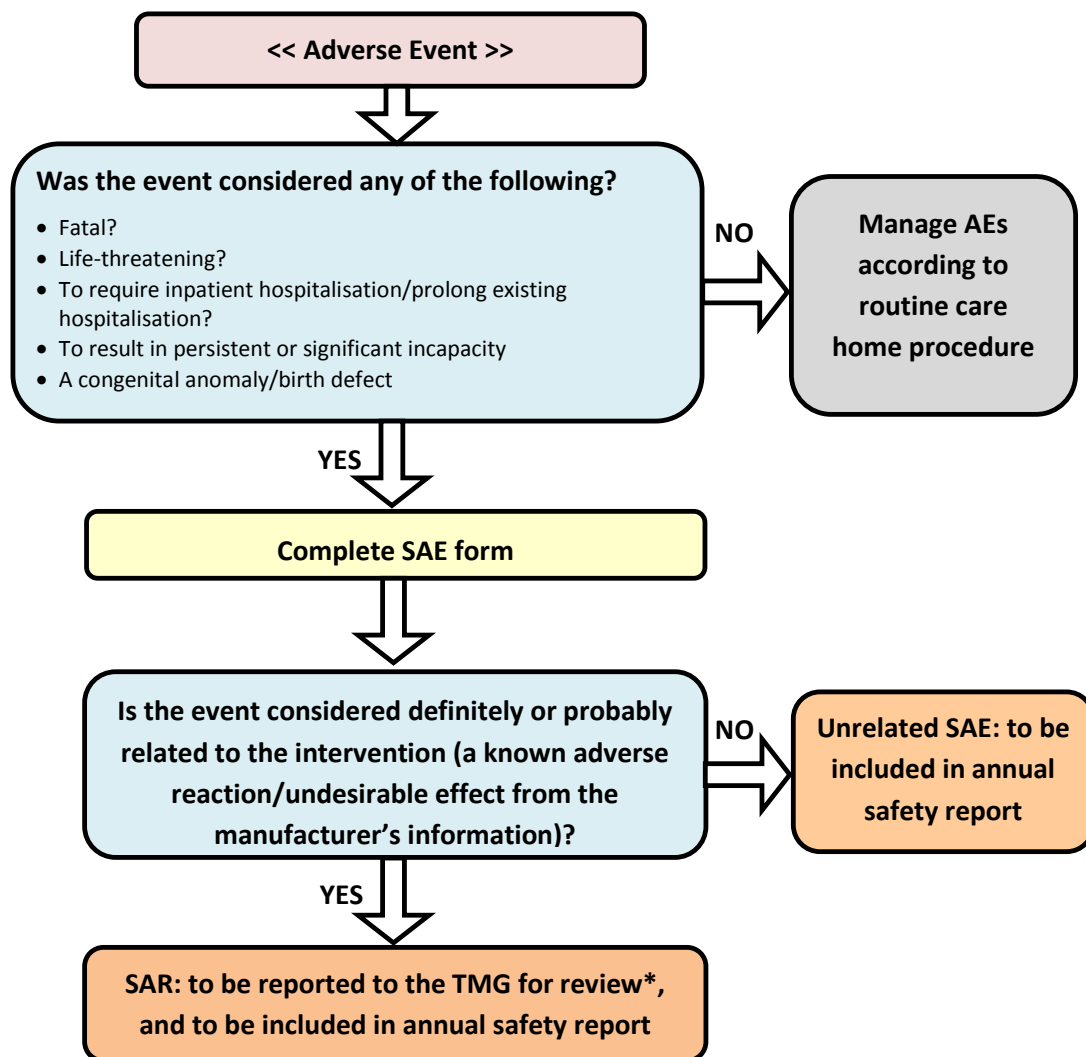
Relationship	Description
Unrelated	There is no evidence of any causal relationship with the trial/study or intervention
Unlikely	There is little evidence to suggest there is a casual relationship (e.g. the event did not occur within a reasonable time after intervention) with the study/trial or intervention. There is another reasonable explanation for the event (e.g. the participant’s clinical condition, other treatment).
Possible	There is some evidence to suggest a causal relationship with the trial/study or intervention (e.g. because the event occurs within a reasonable time after intervention). However, the influence of other factors may have contributed to the event (e.g. the participant’s clinical condition, other treatments).
Probable	There is evidence to suggest a causal relationship and the influence of other factors is unlikely.

Definite	There is clear evidence to suggest a causal relationship and other possible contributing factors can be ruled out.
Not assessable	There is insufficient or incomplete evidence to make a judgement of the causal relationship.

13.3 Reporting procedures

Based on an extensive review of the literature the CI has concluded that probiotics are generally well tolerated by the elderly and the risks to CHR associated with the probiotics used in PRINCESS are very low. In addition, given the potential frailty of the trial population and the high incidence of hospitalisation and death in the course of routine care, we would aim to not cloud any true emerging safety profile by collection of unrelated data. The following reporting procedure should be used for PRINCESS. Any queries concerning adverse event reporting should be directed to the PRINCESS Trial Manager in the first instance.

13.4 SAE flowchart



* TMG will monitor and review against expected health events for this population, and if any untoward patterns emerge, this will be discussed, escalated and actioned accordingly.

CHR Hansen will be notified on an annual basis of any SARs.

The emerging safety data will be reviewed regularly by the Independent Data Monitoring Committee who will be asked to offer an opinion on the potential relatedness to the intervention and the need for any corrective action.

Contact details for reporting SARs

PRINCESS Trial Manager Tel: 029 2068 7601 or Fax: 020 3107 0875

14 Trial procedures

14.1 Location

All trial appointments will take place at the care home where the participant is resident. All trial procedures will be carried out by either the participants' normal care providers or by RNs.

14.2 Data collection/assessment

Eligible participants or their consultee will have an appointment arranged at which informed consent or a consultee declaration (see section 10.3) will be obtained. A second baseline appointment will be made, at which baseline samples will be taken, and baseline data recorded.

Participants will have an appointment arranged to attend a maximum of three additional appointments over a 12 month period. All participants will have two follow ups, one at three months and one at 12 months. A sub-set of patients who have been taking the study product for a minimum of two months and who have had an influenza vaccination after the two month period will have an appointment arranged to attend a third appointment about four weeks (28 days) after the influenza vaccine.

Influenza vaccinations will be administered via routine care. The date of administration will be recorded.

14.3 Trial samples

All trial participants: All trial participants will be asked to provide blood samples at baseline - a full blood count and vitamin D levels will be measured for each participant - the volume of blood taken will be no more than 10ml. At 12 months a full blood count will be measured for 100 participants - the volume of blood taken will be no more than 4ml.

Immunology: For a sample of 100 trial participants some additional immunology work (detailed in section 5.1) will be done and an additional blood sample will be taken at baseline and 12 months – the volume of blood taken will be no more than 6ml at each timepoint. The sample will be taken at the same time as the other blood samples.

Flu vaccine response sub-study: A sub-set of patients will be asked to provide a blood sample on the day of (or up to 10 days prior to this), and four weeks (28 days) after they receive their routine seasonal influenza vaccination. For those agreeing to take part in this flu vaccine response sub-study, 5ml of blood will be taken on the day of, and four weeks after receipt of the influenza vaccination, giving a total of no more than 10ml of blood collected.

Blood samples will be taken by the RN who, if needed, will have undergone geriatric phlebotomy training, or by a care home staff member if for example, they are already taking a blood sample for routine care, and/or the care home resident prefers this. The total volume of blood collected during the 12 month duration of the trial on any one participant will be no more than 36ml.

Each participant will be asked to provide a stool sample at baseline, three months and 12 months. The participants normal care provider will be asked to collect the stool samples.

Each participant will be asked to provide a saliva samples (saliva sample or oral rinse) at baseline, three months and 12 months.

If participants prefer not to provide any or all of the trial samples listed above this will not be a barrier to their entry to the PRINCESS trial.

All samples will be labelled with the participant unique trial identification number before being sent for analysis. Only fully trained and authorised members of the research team and laboratory staff will have access to the samples. The CI will ensure that the storage, analysis and disposal of all clinical samples will meet the requirements of the Human Tissue Act, 2004.

All tissue samples donated by trial participants will be taken on the understanding that they may be used for future research projects in the UK. The samples will be stored and subsequently distributed to approved projects in accordance with the Human Tissue Act and ethical legislation.

Any blood/saliva/stool test results will not be reported back to the CHR GP as they are being collected only for research purposes and will not be analysed in real-time.

14.4 Follow-up

The three and 12 month follow-up visit is detailed above in section 14.3.

15 Statistical considerations

15.1 Randomisation

Participants will be individually randomised in a 1:1 ratio using minimisation. In order to achieve a balance of probiotic and placebo allocations within care homes, the care home from which a participant is recruited will be used as a minimisation variable. A random element, set at 80%, will be used in order to maintain the integrity of the minimisation process. The participants, treating clinicians, trial statistician and research nurses will remain unaware of the group to which participants

are allocated during the trial. Pack IDs will be assigned probiotic / placebo allocations using randomly permuted blocks, and this list will be uploaded into the online system. This process will be carried out by a statistician independent of the trial team.

For more details, please consult the PRINCESS randomisation protocol.

15.2 Sample size

Primary Outcome

A previous study by the research team of antibiotic prescribing in CHR over 12 months estimated an average CAPD of 17.4 days per resident-year. The PRINCESS trial will recruit 330 participants (165 per arm) to allow 90% power at the 5% significance level to detect a 10% relative reduction in CAPD in the probiotic arm (absolute reduction to 15.6 days per resident-year, equivalent to 90 days reduction in CAPD per 50 bed home /year). We consider that a 10% reduction is feasible and would be clinically important. The sample accounts for a 30% drop-out rate. In a previous study of antibiotic prescribing in CHR approximately 30% of participants withdrew or died during the study. However, the median duration of follow-up for all participants (including drop-outs) was 310 days.

Secondary Mechanistic Outcomes

Previous research has found a 40% prevalence of multi-drug resistant *E. coli* in faecal samples of UK nursing home residents (3) and a 37% prevalence of oral candida in hospitalised elderly patients (43). A meta-analysis of 6 trials of probiotics in critically ill patients reported probiotics reduced colonisation with multi-drug resistant gram negatives (OR 0.39 (95% CI: 0.16 to 0.95) [10]. Despite high prevalence of AMR colonisation in CHR few studies have measured the effect of probiotics on this outcome. Hatakka et al (2006) found probiotics reduced the risk of oral candida in 276 older people by 75% (OR = 0.25, 95%CI 0.10- 0.65) (44). Stool and saliva samples at 12 months will provide 90% power at the 5% level to detect a 19% absolute reduction antimicrobial resistant bacteria and oral candida, assuming a 30% drop-out rate.

16 Analysis

16.1 Main analysis

The primary analysis will be by intention-to-treat, and will consist of a between-group comparison of the mean cumulative antibiotic administration days (CAAD) using Poisson regression. As the randomisation will be stratified by care home, the regression model will control for the care home a participant was recruited from. Negative binomial regression will be used in the presence of over-dispersed count data.

For secondary outcomes, depending on the type of data, a mixture of Poisson, linear, logistic and Cox models will be used to appropriately compare trial arms with respect to rates, means, proportions and time to events (see table below for more details).

Adherence-adjusted analysis

While our primary analyses will be conducted using the intention-to-treat principle (i.e. a comparison of groups as randomised regardless of what happens after randomisation), we will monitor adherence to study product throughout the trial. Using this data, we will perform adherence-adjusted analyses, deriving estimates of treatment efficacy that maintain a comparison of groups as randomised, using structural mean models (39). A definition of adherence to study product will be defined prior to the commencement of any statistical analysis.

Missing data assumptions and adjustments

Given our proposed intensive monitoring schedule, missing data is likely to be minimal for participants who remain in the trial for the full duration. Where missing data is likely to occur, it will most likely be due to participant drop-out, with reasons for dropout falling into two broad categories, withdrawal from the trial and death.

Where responses are missing due to drop-out, these will be assumed to be missing at random given observed data, and appropriate modelling techniques will be used (e.g. likelihood-based methods or multiple imputation). Sensitivity analyses will be conducted using joint modelling approaches (e.g. selection and/or pattern mixture models) to explore departures from the missing at random assumption (40).

Survivor Average Causal Effects (SACE) will also be estimated. These provide estimates of treatment effects in the subgroup of participants who would remain alive regardless of the group to which they were randomised. This method acknowledges that censoring after randomisation due to death may be a process that is subject to unmeasured selection bias, and uses a principal stratification framework to calculate a treatment effect based on potential outcomes (41).

We will also explore stratifying deaths into two groups: death during an infection and death from any other cause. We will then investigate applying a missing at random-valid modelling approach to the deaths from any other cause and take a more extreme approach with those that died during an infection (e.g. assume that they would have remained on antibiotics for the remainder of the follow-up period).

Similarly, we will explore stratifying drop-out due to residents moving to another care facility into two groups: those who move for health-related reasons (e.g. from a residential home to a care home providing nursing care) and those who move for other reasons. A missing at random-valid modelling approach will be applied to the latter strata, with a more extreme approach taken for those dropping out due to moving from the care home for health-related reasons as this represents a decline in health, which may also be subject to unmeasured selection bias.

Analysis for mechanistic process

Further statistical modelling will explore the causal mechanisms by which the probiotic may have an effect. Mediation analyses will explore the effect of exposure to probiotics on CAAD and cumulative number of infection days, and this is mediated through an effect on antimicrobial resistance. These analyses will be performed using G-computation (43, 44).

16.2 Description of outcomes and method of statistical analysis

Outcome	Main analysis	Sensitivity analyses*
Primary		
Cumulative antibiotic administration days (CAAD)	Poisson regression.	Negative binomial regression (if Poisson over-dispersed).
Secondary		
CAAD by infection (UTI, RTI, skin, GI, unexplained fever)	Poisson regression, with trial arm interacted with infection.	Negative binomial regression (if Poisson over-dispersed).
Incidence of infection	Poisson regression.	Negative binomial regression (if Poisson over-dispersed).
Site of infection	Poisson regression, with trial arm interacted with infection.	Negative binomial regression (if Poisson over-dispersed).
Mean duration of infection	A two-level Cox PH (frailty) model with infections nested within participants.	
Cumulative number of infection days	Poisson regression.	Negative binomial regression (if Poisson over-dispersed).
Prophylactic antibiotic use (cumulative prophylactic antibiotic administration)	Poisson regression.	Negative binomial regression (if Poisson over-dispersed).
Secondary – Health Utility		
Health utility (EQ5D)	Mixed linear regression (EQ5D at 3/12 months nested within participants) controlling for baseline EQ5D.	Transformations for normality.
Well-being (ICECAP-O)	Mixed linear regression (ICECAP-O at 3/12 months nested within participants) controlling for baseline ICECAP-O.	Transformations for normality.
GP visits	Poisson regression.	Negative binomial regression (if Poisson over-dispersed).
A&E visits	Logistic regression.	Poisson/negative binomial regression of counts (mean rate of A&E visits).
Hospitalisations	Logistic regression.	Poisson/negative binomial regression of counts (mean rate of A&E visits).
Death	Logistic regression.	Cox PH model for time from

		randomisation to death.
Secondary - Antimicrobial Resistance		
Presence of Gram-negative isolates in stools	Mixed logistic regression (stool sample finding at 3/12 months nested within participants) controlling for baseline stool sample finding.	Interaction between time point and trial arm.
Vancomycin resistant enterococci in stools	Mixed logistic regression (stool sample finding at 3/12 months nested within participants) controlling for baseline stool sample finding.	Interaction between time point and trial arm.
Secondary - Mechanistic		
Immune parameters -TLR ligand stimulated cytokines/ cytokines/chemokines -PBMC TLR expression -Flu vaccine response	Logistic regression.	
<i>Clostridium difficile</i> in stools	Mixed logistic regression (stool sample finding at 3/12 months nested within participants) controlling for baseline stool sample finding.	Interaction between time point and trial arm.
Presence of oral candidiasis	Mixed logistic regression (response at 3/12 months nested within participants) controlling for baseline response.	Interaction between time point and trial arm.
Amount of oral candidiasis present in saliva	Mixed linear regression (response at 3/12 months nested within participants) controlling for baseline response.	Interaction between time point and trial arm.
<i>L. rhamnosus</i> GG in stools	Mixed logistic regression (stool sample finding at 3/12 months nested within participants) controlling for baseline stool sample finding.	Interaction between time point and trial arm.

* All sensitivity analyses will include a consideration of adherence to study product and the impact of missing data / deaths.

16.3 Data storage & retention

Access to Data

Direct access to trial data will be granted to authorised representatives from the sponsor or host institution for monitoring and/or audit of the trial to ensure compliance with regulations. This access, the reason for it and who has authorised it will be recorded by the trial team and reported to the DMC. Only authorised members of the PRINCESS research team will have access to the PRINCESS trial data in order to carry out their assigned trial role. Who has access, what level of access and for what

purpose will be recorded in the trial Data Management Plan. All data will be kept for 15 years in line with Cardiff University's Regulations for clinical research. This data will be stored confidentially on password protected servers maintained on the Cardiff University Network.

Data Recording and Record Keeping

A trial specific Data Management Plan (DMP) will be developed for the PRINCESS trial outlining in detail the trial specific procedures that will be put in place to ensure that high quality data are produced for statistical analysis. All data recorded onto paper Case Report Forms (CRFs) or other paper data sources will be returned to the trial team and a copy will be held at the research site. Upon receipt all CRFs or other paper documentation containing clinical data will be date stamped and tracked until archiving. A full pre-entry review and electronic data validation for all data entered into the clinical database will be provided by trial specific programmed checks. Prior to database lock, a dataset review will be undertaken by the Data Manager and the Trial Statistician. An independent review of the quality of the data being produced will also be provided by the IDMC throughout the trial.

17 Trial closure

The end of trial is the date of the last patient recruited into the PRINCESS trial has completed the intervention period of 12 months, all follow-up data has been collected and laboratory analysis completed.

18 Regulatory issues

18.1 Ethical and research governance approval

The study will be conducted in accordance with the recommendations for physicians involved in research on human participants adopted by the 18th World Medical Assembly, Helsinki 1964 as amended.

The protocol, informed consent form, participant information sheets, consultee declaration form, consultee information sheet and any proposed advertising material will be submitted for written approval to an appropriate Research Ethics Committee (REC); host institution(s); and for Research governance approval, prior to any trial procedures taking place. The Health Research Authority guidance on consent and participant information sheets will be followed as appropriate. The Chief Investigator (CI) or delegate will submit and, where necessary, obtain approval from the above parties for all substantial amendments to the original approved documents.

18.2 Consent

The PRINCESS trial will recruit participants both with capacity and those lacking capacity to provide informed consent. We anticipate that some care home residents may lose capacity during the course of the trial either temporarily or permanently and will make appropriate provision. Specific consent will be sought at trial entry for CHR to remain in the trial should capacity be lost during the trial and where CHR regain capacity, consent will be subsequently obtained. The CI will ensure that the consent process adheres to the Mental Capacity Act, 2005 as appropriate.

Full study specific training will be available to all care home staff. Only care home staff who have undertaken the appropriate training for their trial role will take responsibility for taking informed consent or other trial procedures. The staff along with their trial role will be recorded in the trial delegation log and authorised by the CI and care home representative. The type of training needed and how it will be recorded will be noted as a trial specific risk and reviewed as part of the risk assessment process.

18.3 Risks and benefits

Due to this being a randomised placebo controlled trial, half of the participants have no chance of benefit from the trial intervention. All participants (or their consultee), however, may value contributing to the advancement of medical knowledge and appreciate the increased assessments and monitoring. Probiotics are classed as a food supplement (and this has been confirmed by the MHRA for this trial) and there are very few side effects (mainly bloating and flatulence if these occur). Probiotics carry theoretical risks including infection beyond the gut, and transfer of antibiotic resistant genes. However, so far, there have been no reports of bacteraemia or fungaemia attributable to the probiotics in trials included in the systematic reviews and mild to moderate gastrointestinal side effects and rash are generally no more common than in patients on placebo probiotic.

For participants receiving the probiotic, reducing the possibility of developing an infection that may require treatment with an antibiotic, we believe, outweighs the risk of the potential side effects. If our hypothesis is correct and probiotics reduce antibiotic prescribing for CHR, this may also reduce the number of antibiotics prescribed for other CHR and subsequent AMR. For all participants, the high level of monitoring for common infectious diseases may result in better-targeted treatment and added vigilance in general.

Residents lacking capacity will be included in the trial as our previous PAAD observational study data demonstrated that those who lack capacity are frailer and receive antibiotics more often and for different indications than those with capacity and, therefore, may be more likely to benefit from interventions aimed at stimulating and supporting the immune system. Interventions should be evaluated in populations for whom the intervention is intended, especially when there are relevant material differences (e.g. in frailty and in immune function).

18.4 Confidentiality

The CI will ensure all trial staff are fully trained and adhere to the principles of Good Clinical Practice (GCP) and the Data Protection Act, 1998. Patients will only be identified on trial documents by use of a unique trial ID which cannot be used to identify individual participants. The research team will store all CRFs and other trial data documents securely at the coordinating centre prior to data entry. CRFs and all other documents holding patient identifiable information will be anonymised as soon as possible with the process of management being outlined in detail within the ethics application and in the Data Management Plan.

18.5 Indemnity

Cardiff University will provide indemnity and compensation in the event of a claim by, or on behalf of participants, for negligent harm as a result of the study design and/or in respect of the protocol authors/research team. Cardiff University does not provide compensation for non-negligent harm.

All participants will be recruited at Care Homes. Care Homes will arrange insurance cover for any harm to participants caused by the negligence of the Care Home.

18.6 Trial sponsorship

Cardiff University will act as sponsor for trial. Delegated responsibilities will be assigned to SEWTU, PCCTU and to the care homes taking part in this study.

18.7 Funding

The trial will be funded by Medical Research Council's Efficacy and Mechanism Evaluation Programme administered by the National Institute for Health Research by means of a research grant to the Nuffield Department of Primary Care Health Sciences. The research funding will be administered by the University of Oxford. The study product (probiotic/placebo) will be provided by CHR Hansen free of charge.

18.8 Audits & inspections

The trial is participant to inspection by National Institute for Health Research Efficacy and Mechanism Evaluation as the funding organisation. The trial may also be participant to inspection and audit by Cardiff University under their remit as sponsor.

19 Trial management

The Trial Management Group will be led by the Chief Investigator and will take responsibility for the on-going management of the PRINCESS trial. Members will be required to sign up to the remit and conditions as set out in the TMG Charter.

20 Data monitoring & quality assurance

The UKCRC accredited South East Wales Trials Unit (SEWTU) will have fully responsibility for all aspects of the PRINCESS trial. SEWTU will work closely with the Primary Care Clinical Trials Unit (PC-CTU) on all matters relating to the trial design and delivery. A formal division of responsibilities will be drafted between SEWTU and PC-CTU detailing which standard operating procedures will be followed when completing each controlled process. SEWTUs Quality Assurance Team will review these arrangements as appropriate to ensure compliance with the relevant regulations and any Sponsor requirements.

20.1 TSC (Trial Steering Committee)

An independent Trial Steering Committee (TSC) will be established to provide oversight of the PRINCESS trial. The TSC will include at least an independent chairperson, two independent members and a patient representative. The TSC will review the progress of the trial on a regular basis and provide advice to the TMG. Representatives of the Sponsor and the Funder will be invited to attend all meetings. Members will be required to sign up to the remit and conditions as set out in the TSC Charter.

20.2 IDMC (Independent Data Monitoring Committee)

An independent Data Monitoring Committee (DMC) will be established to provide oversight of all matters relating to patient safety and data quality. The DMC will be asked to convene at least annually and provide advice to the TSC. Members will be required to sign up to the remit and conditions as set out in the DMC Charter.

21 Publication policy

All publications and presentations relating to the trial will be authorised by the Trial Management Group and will follow the PRINCESS publication policy. The trial protocol will be published and the trial registered with the ISRCTN. The trial results will be published and all who meet the criteria for authorship will be listed as authors. Authorship will be determined in accordance with the ICMJE guidelines and other contributors will be acknowledged. The results will be presented to the DMC prior to publication. Papers will be shared with the funders prior to submission. Funders will have 14 days in which to respond and to bring any matters of factual accuracy relating to the intervention to the attention of the trial team. The funders will have no role in decisions on publication. The funding source and other support will be acknowledged.

21.1 Feedback to participants and other stakeholders

Participants, care home staff and associated GPs will receive summaries of the trial findings and they will be made available to the general public via the trial website. The PPI representatives will be asked for their assistance in ensuring the material prepared for the care home residents and general public is comprehensive and appropriate.

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23 Schedule of Study Procedures

Assessment	Baseline	Once Daily (for 12 months)	Regular intervals (for 12 months)	Monthly (for 12 months)	If vaccinated against flu, on day of vaccination or as close as possible	If vaccinated against flu, 4 weeks (28 days) afterwards	3 month	12 month
Assessment of eligibility	X							
Informed consent or consultee declaration*	X							
Demographic data	X							
Brief clinical history	X							
Clinical Frailty Score	X							
CAAD for infection			X				X	X (from MAR and hospital record, if applicable)
Infection details			X				X	X
Oral rinse or saliva culture*^ for oral candida	X						X	X
Blood sample*^ (FBC, Vitamin D)	X							
Blood sample*^ for serum cytokines (ex-vivo response) (n≈100)	X							X plus FBC blood sample
Blood sample*^ (influenza vaccine efficacy)					X	X		
Stool sample*+ (C.diff, antibiotic sensitivity of gram -ve enterobacteria and VRE, gut microbiota diversity or phylogroup, microbiome comp.)	X						X	X
EQ-5D* (participant and/or proxy completion)	X						X	X
ICECAP-O* (participant and/or proxy completion)	X						X	X
NHS service use (Care Home note review)			X					X
NHS service use (Hospital note review)								(X)
Mortality								X
Randomisation	X							
Study product pack allocation				X				
Study Product administration		X						
Adherence to Study Product (MAR sheet)		X (MAR sheet completion as often as practicable)	X (transcri be to CRF)					
Adherence to Study Product (Clinical Rating scale)				X				
Adherence to study product (capsule count)				X				

Key

- * requires participant or consultee/proxy involvement
- (X) if applicable
- + Sample to be collected by Care Home staff
- ^ Sample to be taken by Research Nurse

Follow-up time windows:

- 3 month:** 3 months +/- 2 weeks
- 12 month:** 11 months +/- 1 month

Short title: PRINCESS SAP

Statistical Analysis Plan for The PRINCESS Study

Full study title: A double-blind placebo controlled trial to evaluate the efficacy of probiotics (*Lactobacillus rhamnosus*, LGG and *Bifidobacterium animalis* subsp. *lactis*, BB-12) in reducing antibiotic administration for infection in care home residents

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Draft Plan

This document has been written based on information contained in the study protocol version v3.0, dated 25 April 2018

SAP Revision History

Protocol version	Updated Sap version no.	Section number changed	Description and reason for change	Date changed

Short title:	PRINCESS SAP
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ROLES AND RESPONSIBILITIES

Trial Statistician: Dr David Gillespie			
Role: Author of the SAP			
Date:	29/10/2018	Signature:	
Trial Statistician: Miss Mandy Lau			
Role: Author of the SAP			
Date:	12/12/2018	Signature:	
Senior Statistician: Professor Kerenza Hood			
Role: Review and approve			
Date:	29/10/18	Signature:	
Chief Investigator: Professor Christopher Butler			
Role: Review and approve			
Date:	14/01/19.	Signature:	

Short title: PRINCESS SAP

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1. INTRODUCTION

This statistical analysis plan provides guidelines for the final presentation and analysis for the PRINCESS trial. This plan, along with all other documents relating to the analysis of this trial, will be stored in the Statistical Analysis Master File electronically and/or in hard signed copy formats.

2. BACKGROUND

2.1 RATIONALE AND RESEARCH QUESTION

Care home residents (CHR) are prescribed far more antibiotics than the general population(1). In our previous research in care homes, we found that CHR took antibiotics for an average of 17.4 days per year(2). High antibiotic use increases the risk of reservoirs of antimicrobial resistance (AMR) in care homes that can spread within care homes and to hospitals and the community(1). Reviews of studies found that probiotic reduced antibiotic use and risk of respiratory tract infections in adults(3) and reduced antibiotic-associated diarrhoea(4). Probiotics also reduced carriage of AMR bacteria in a review of critically ill patients, and have reduced duration of common infectious diseases(5) and enhanced immune response, including to flu vaccination(6). However, research in CHR is currently lacking.

2.2 OBJECTIVES

2.2.1 RESEARCH HYPOTHESIS

The null hypothesis for our primary analysis is that there is no difference between care home residents (CHR) given a daily probiotic for up to 12 months and those given a daily placebo for up to 12 months in their cumulative antibiotic administration days (CAAD) for infections. The alternative hypothesis is that there is a difference between the two arms.

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2.2.2 STUDY OBJECTIVE

The primary objective of this trial is to evaluate the effect of a daily dose of oral probiotics on CAAD for all-cause acute infections.

The secondary objectives of this trial are to determine the effect of daily probiotic intake on:

- a) CAAD for the following sub-categories of infection: respiratory tract infections, urinary tract infections, skin infections, gastrointestinal infections, unexplained fever and other infection
- b) Site, incidence and duration of infection
- c) Prevalence of *C. difficile* infection
- d) Health Utility (EQ-5D-5L(7))
- e) Wellbeing (ICECAP-O(8))
- f) Service Use (hospitalisation)
- g) Mortality
- h) Antibiotic associated diarrhoea

3. STUDY MATERIALS

3.1 TRIAL DESIGN

The PRINCESS trial is a two-arm double-blind individually-randomised controlled trial which will take place approximately across 20 care homes in Wales and England. The trial will evaluate the efficacy and mechanisms of a nutritional intervention (daily dose of oral probiotics) on cumulative antibiotic administration days for all-cause infections and incidence and duration of infections in care home residents over a 12 month period. Treatment allocation is a 1:1 ratio. Care home residents are randomised to either daily probiotic supplement or matched placebo for up to a 12-month period.

3.2 RANDOMISATION

The researcher (either data manager, research nurse, or trial manager) will randomise the participants on an individual basis using the online computerised randomisation system (Sortition) provided by the University of Oxford Primary Care Clinical trials Unit. The system will be operational 24 hours a day. Allocation ratio will be one-to-one and stratified by care home, and minimised by gender of participant. The algorithm will allocate new participants to the trial arm that minimises imbalance with respect to gender and within care homes with probability of 0.8. Full details of the randomisation process are described in the trial protocol and randomisation protocol.

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3.3 SAMPLE SIZE

The original target for the PRINCESS trial was to randomise 330 participants from around 20 care homes in Wales and England. This would provide 90% power at the 5% level to demonstrate a 10% relative reduction in cumulative antibiotic administration days (CAAD) (assuming an average CAAD of 17.4 days and an absolute reduction in the probiotic arm to 15.6 days per resident-year(2)). We consider that a 10% reduction is feasible and would be clinically important.

This sample size accounted for 30% of participants being lost to follow-up due to withdrawal or death during the study.

An interim assessment of primary outcome ascertainment revealed that the mean percentage of days for which there are valid antibiotic administration data (i.e. either no antibiotics administered, or the number of days antibiotics have been administered in a given week) is 77.4%, or 283 days out of a possible 365 on average. This percentage varies depending on the length of time participants have been in the study. However, it remains high for participants who have been in the study for over six-months, and does not take into account other data sources that might be used (e.g. hospital discharge summaries, medication administration records, etc.). This is likely to therefore reflect an underestimate of the availability of primary outcome data, but nevertheless is a more accurate reflection of the likely level of follow-up when compared to our original assumption.

Given slower than anticipated recruitment, and this new information regarding the trade-off between the number of participants required and average length of follow-up, we will aim to randomise between 258 and 270 participants. Assuming a mean number of days for which primary outcome data will be available (i.e. accounting for follow-up time and missing data) of approximately 250 days, this will provide at least 82% power to detect a 10% relative reduction in CAAD.

3.5 INTERIM ANALYSES

No interim analyses are planned.

3.6 TIMING OF FINAL ANALYSIS

Final analysis for treatment arm comparison is planned to take place when every participant has reached the final follow-up (or is no longer participating in the study) and data for all CRFs and collection forms have been received and cleaned.

3.7 TIMING OF OUTCOME ASSESSMENT

The schedule of study procedures is given below and also outlined in Appendix I:

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- Screening: Screening will be conducted prior to recruitment to assess participant eligibility and obtain consent
- Baseline: Baseline data will be collected following consent but prior to randomisation
- Randomisation: Following baseline assessments but prior to a study product being allocated to the CHR
- Weekly review: Weekly participant record collected every week during the 12-month follow-up period by research nurses
- Three-months follow-up: Face-to-face data collection approximately three-months post-randomisation
- 12-months follow-up: Face-to-face data collection approximately 12-months post-randomisation*
- Flu vaccine response sub-study: Blood sample will be taken on the day of (or up to 10 days prior to), and four weeks (28 days) post influenza vaccination. Participants need to have been on study product for at least 2 months prior to influenza vaccination. **[NB. Further detail of this sub-study is provided in the mechanistic SAP].**

The outcomes that are collected at each visit are given in Appendix 1 and described in full within the trial protocol.

*Due to slower than anticipated recruitment, we are truncating the second face-to-face follow-up for some participants. The minimum time of the second face-to-face follow-up will be six-months post-randomisation. The time at which this second follow-up visit takes place will be accounted for in any analysis involving this time point.

4. STATISTICAL PRINCIPLES

4.1 LEVELS OF CONFIDENCE AND P-VALUES

All confidence intervals presented will be 95% and two-sided. All applicable statistical tests will be two-sided and will be performed using a 5% significance level.

4.1.1 ADJUSTMENT FOR MULTIPLICITY

No adjustments will be made for multiple testing.

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4.2 ADHERENCE AND PROTOCOL DEVIATIONS

4.2.1 DEFINITION AND ASSESSMENT OF ADHERENCE

Data regarding participants' adherence to the study product will be collected from a number of sources:

- MAR sheets - administration of the study product will be recorded by care home staff using Medication Administration Record sheets (MAR sheets) and monitored by RNs during their visits to each care home.
- Presence of probiotic organisms in bowel - the presence of the probiotic organisms will be assessed in stool samples of participants at baseline, three months and final follow-up.
- Capsule count - counts of unused study product will be undertaken by the PRINCESS trial team at the end of the study for each participant (i.e. final follow-up, withdrawal, or death).

4.2.2 PRESENTATION OF ADHERENCE

Descriptive statistics will be reported both overall and split by trial arm.

Our primary measure of adherence will be based on MAR sheets, as it is the only measure that is able to provide data regarding patterns in adherence over time (i.e. daily consumption and time of consumption). Capsule counts and presence of probiotic organisms in the bowel are measures that will be used to verify adherence data obtained from the MAR sheets. Exploratory analyses will consider the agreement between the different types of measures.

For each participant, the total number of days they took the study product will be expressed in a summary form as a percentage, with the denominator being the total number of days they were observed (i.e. still taking part in the trial, regardless of whether or not they had withdrawn from the study product).

This will be summarised both overall and between trial arms as means and standard deviations or medians and interquartile ranges, as appropriate.

Adherence will be also quantified as three distinct processes, using the ABC taxonomy(9):

Initiation: A participant will have been deemed as having initiated their Study Product if the data collected from MAR sheets indicates they used it at least once during the follow-up period. This will be summarised as frequencies and percentages.

Implementation: In those that initiate their Study Product, implementation will be defined as the percentage of days that a participant takes their Study Product as intended (i.e. once daily for the

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duration of the follow-up period). This will be summarised as means and standard deviations or medians and interquartile ranges, as appropriate.

Daily implementation, whether Study Product was taken will also be described and modelled over time (see 6.4.2 for more details).

Discontinuation: A participant will be defined as having discontinued use of their Study Product if they withdraw from the Study Product component of the trial (but remain participating in other aspects of the trial). The day after they were last recorded as having taken the Study Product will be defined as the day they discontinued. This will be summarised as medians and interquartile ranges and graphically using Kaplan Meier plots.

The extent to which adherence (in both its summary form and individual elements) is clustered by care home will be investigated and reported.

4.2.3 DEFINITION OF PROTOCOL DEVIATION

The definition of protocol deviation is any change, divergence, or departure from the study design or procedures defined in the protocol. Important protocol deviations are a subset of protocol deviations that may significantly impact the completeness, accuracy, and/or reliability of the study data or that may significantly affect a subject's rights, safety, or well-being. Protocol deviations of relevance (and that will not be identified under other sections, such as adherence or missing data issues) for the PRINCESS study include:

- Inclusion of an individual not meeting eligibility criteria;
- Administering of incorrect study product to a participant (i.e. study product with another individual's ID).

4.2.4 PRESENTATION OF PROTOCOL DEVIATIONS

Protocol deviations will be reported for all participants and separately for each trial arm and by site. The frequency of any protocol deviation and frequency of different types will be reported.

4.3 ANALYSIS POPULATION

Primary and secondary comparative analyses will be conducted on an intention-to-treat (ITT) basis without imputation of missing values (complete case set). The ITT population uses all randomised participants in the arms they were randomised to regardless of the intervention received and the complete case population (CC) restricts the population to include all randomised participants with complete follow-up (specific to the outcome and time points).

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ITT analysis with missing values imputed will be undertaken as a sensitivity analysis (see Section 6.3 for more details).

To study the effect of receiving probiotics (rather than just being randomised to them), we will use instrumental variables methods, (10) with randomisation as the instrument, and two measures of exposure:

1. Initiation (Y/N) – this will provide an estimate of the effect of (at least) initiating probiotics;
2. Implementation (as a continuous variable) – using the definition in 4.2.2, with non-initiators set to zero, this will provide an estimate of the effect on outcome per percentage point increase in probiotics use

5. STUDY POPULATION

5.1 SCREENING DATA

A screening log of all ineligible and eligible but not consented/not approached will be kept at each care home so that any biases from differential recruitment (with respect to age and gender) will be detected.

5.2 ELIGIBILITY

Summary statistics on eligibility, recruitment, withdrawal and dropout will be collated for both trials arms and will form the basis of the CONSORT flow diagram for clinical trial reporting(10).

5.3 RECRUITMENT

A CONSORT diagram will be used to summarise the number of CHR who were:

- Assessed for eligibility at screening
 - eligible at screening
 - ineligible at screening*
- eligible and randomised
- eligible but not randomised*
- received the randomised allocation
- did not receive the randomised allocation*
- agreed to provide stool/saliva/blood samples
- provided stool/saliva/blood samples at baseline/3 months/final follow-up (from those agreed to provide samples)

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- stool/saliva/blood samples processed at baseline/3 months/final follow-up (from those who provide samples)
- lost to follow-up at 3 months and final follow-up*
- discontinued the intervention*
- randomised and included in the primary analysis
- randomised and excluded from the primary analysis*

*reasons will be provided

Appendix II provides an example of the CONSORT diagram that will be used.

5.4 WITHDRAWAL/FOLLOW UP

5.4.1 LEVEL OF WITHDRAWAL

The level of consent withdrawal will be tabulated and classified as following:

- Withdrawal from entire trial and does not want any data or samples already collected relating to them to be used
- Withdrawal from trial, study product and all subsequent trial follow-up (sample collection and data collection), but data and samples already obtained up to this point can be used
- Withdrawal from study product but happy to continue with follow-up processes (to include data collection and/or sample collection) – this may be due to a request to withdraw, or as a result of a change in the participant's condition or circumstances which justifies the discontinuation of the study product
- Withdrawal from study product and all subsequent sample and questionnaire data collection, but happy for routine care record data to be collected.

5.4.2 TIMING OF WITHDRAWAL

Participant may withdraw consent to participate in this study at any time. Information on the timing of withdrawal will be presented in the CONSORT diagram, with numbers and reasons for withdrawal, in relation to the key study time points (after screening, before baseline, before randomisation, after three months follow-up, etc.).

5.4.3 REASONS FOR WITHDRAWAL

Reasons for withdrawal will be tabulated and compared between trial arms to ascertain any differential bias that may be present in those who have remained in the study.

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5.4.4 PRESENTATION OF WITHDRAWAL/LOSS TO FOLLOW-UP

Summary statistics on participant assessment, eligibility, recruitment, withdrawal and death will be collected for both treatment arms and form the basis of the CONSORT diagram.

5.5 BASELINE PARTICIPANT CHARACTERISTICS

5.5.1 LIST OF BASELINE DATA

Participants will be described with respect to age, gender, mental capacity, ethnicity, height, weight*, ulna length, mid upper arm circumference, clinical frailty scale, prescribed antimicrobials in the last four weeks and use of other medications in last four weeks (proton pump inhibitor, laxative and vitamin D supplement), both overall and separately for the two trial arms.

*Due to the fluctuating nature of weight in this population, only weight measurements taken up to three-months prior to randomisation will be reported.

5.5.2 DESCRIPTIVE STATISTICS

Participant flow and recruitment

Summary statistics on patient assessment, eligibility, recruitment, withdrawal and dropout will be collated for both trial arms and form the CONSORT flow diagram for clinical trial reporting. Specifically, for each arm, participants randomly assigned, receiving intended intervention, completing the study protocol, and analysed for the primary outcome. Any protocol deviations will be described with reasons as well as dates defining the period of recruitment and follow-up.

Baseline data

Appropriate descriptive summaries and graphical illustrations of baseline demographic, clinical and questionnaire data will be presented (see Appendix III for example table). Descriptive summaries for the minimisation variable, gender, will be produced at site level. Analyses by trial arm will be performed when all follow-up data has been obtained and cleaned. Baseline data will be used to describe balance between trial arms and generalizability of the study population. There will be no formal testing of between-arm differences for any variables at baseline.

Follow-up data (weekly data, three-months and final follow-up)

Further descriptive summaries for the sample as a whole and by trial arm will be tabulated after data collection completed. Graphical illustration (boxplots, histograms and bar charts) will be used where

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appropriate and the distributions of all continuous outcomes will be checked with transformation used as necessary. Rates will be expressed per person year.

6. ANALYSIS

6.1 OUTCOME DEFINITIONS

6.1.1 PRIMARY OUTCOME

Outcome Domain:	Outcome	Time point(s)	Outcome type
Infection	Cumulative antibiotic administration days (CAAD)	Up to 12 months	Continuous

6.1.2 TIMING, UNITS AND DERIVATION OF PRIMARY OUTCOME

Cumulative antibiotic administration days (CAAD) will be collected for up to 12 months post-randomisation. The CAAD will be expressed as a rate, with the number of exposure days (i.e. time at risk) as the denominator and antibiotic administration days the numerator. If a participant has been prescribed an antibiotic but this has not been administered, this will be regarded as not taken and will not count towards the analysis. The focus of this outcome is on antibiotics for systemic infection/s. Therefore, any topical antibiotics or antibiotics (of any route) prescribed as prophylaxis will be excluded from the numerator. Prophylactic antibiotics will either be defined in free-text or defined as antibiotic administration (same type) lasting for > 30 days. These exclusions apply to all CAAD outcomes. Assumptions around CAAD that will be made during periods of hospitalisation are listed in Appendix IV.

6.1.3 LIST OF SECONDARY OUTCOMES

Outcome Domains:	Outcome	Time point(s)	Outcome type
Infection	1. CAAD: cumulative antibiotic administration days on the following sub-categories of infection respiratory tract infections (RTI), urinary tract infections (UTI), skin infections, gastrointestinal infections (GI), unexplained fever and other	Up to 12 months	Count
	2. Incidence of infection (any)		Count
	3. Incidence of different types of infection (RTI, UTI, skin infections, GI and unexplained fever)		Count

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Outcome Domains:	Outcome	Time point(s)	Outcome type
	4. Mean duration of infection		Continuous
	5. Cumulative number of infection days		Count
Health Utility and wellbeing	1. EQ-5D-5L: Self and proxy	Baseline, 3 months and final follow up	Continuous (score from -0.594 to 1)
	2. ICECAP-O : Self and proxy reported ICECAP ability measure for Older people	Baseline, 3 months and final follow-up	Continuous
	3. Ever been hospitalised (Yes/No)	Up to 12 months	Binary
	4. Number of hospital stays	Up to 12 months	Count
	5. Cumulative number of hospital days	Up to 12 months	Count
	6. Death (Yes/No)	Up to 12 months	Binary
Diarrhoea	1. Antibiotic associated diarrhoea (incidence and cumulative days)	Up to 12 months	Binary and Count
	2. All-cause diarrhoea (incidence and cumulative days)	Up to 12 months	Binary and Count
	3. Mean duration of diarrhoea episodes (i.e. mean duration of consecutive diarrhoea days)	Up to 12 months	Binary and Count

6.1.4 ORDER OF TESTING

All analyses will be conducted in series. None are dependent on each other.

6.1.5 TIMING, UNITS AND DERIVATION OF SECONDARY OUTCOMES

The timing and units of secondary outcomes is provided in the protocol and in section 6.1.3 of this document. Key points and principles related to the derivation of secondary outcomes are listed below:

- **CAAD on the following sub-categories of infection respiratory tract infections (RTI), urinary tract infections (UTI), skin infections, gastrointestinal infections (GI), unexplained fever and other:** Where antibiotics are administered in a week where more than one infection type is

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indicated (e.g. RTI and UTI, the CAAD will be counted for both infection types. The frequency of these instances will be reported.

- **Incidence / site / duration of infection:** Where the same infection type (e.g. UTI) occurs with an infection and antibiotic-free gap of at least seven days, this will be counted as two separate infections. See Appendix V for assumptions regarding infections while in hospital.
- **Diarrhoea and antibiotic associated diarrhoea:** Care staff will provide assessments of whether or not a resident has experienced diarrhoea on a given day. Care staff will use routinely collected stool charts to inform this assessment, which contain data on stool consistency and the extent to which the consistency reflects a change from normal habits. Antibiotic associated diarrhoea will be defined as diarrhoea occurring following administration of antibiotics and up to eight weeks after stopping antibiotic treatment. (2) Where episodes of diarrhoea occur with a gap of at least two days this will be counted as two separate episodes of diarrhoea.

6.2 ANALYSIS METHODS

6.2.1 LIST OF METHODS AND PRESENTATION

Primary analysis

The differences in the mean cumulative antibiotic administration days (CAAD) will be compared between trial arms. A two-level Poisson regression model, with responses from care home residents (CHRs) nested within care home, will be fitted in order to evaluate the effect of oral probiotics on CAAD for all-cause infections.

The CAAD count will be included as the outcome. Participant gender will be included in the model, as it was minimised on. Since the observation time for each participant is likely to vary, the length of time observed will be included in the model.

The results will be presented using the mean and standard error for the rate in each trial arm, the adjusted incidence rate ratio (IRR) between trial arms, along with the 95% confidence interval (CI) and p-value. Negative binomial regression will be used in the presence of over-dispersed count data.

Secondary analysis

For the analysis of each of the secondary outcomes, models will include participant gender as a fixed effect and care home as a random effect.

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CAAD by infection (UTI, RTI, skin, GI, unexplained fever)

Separate Poisson regression models will be fitted to explore the treatment effect on CAAD for each infection type (UTI, RTI, skin, GL, unexplained fever). The length of observation time per CHR will be included in the model. The means, standard errors, adjusted IRR will be provided alongside 95% CI and p-values.

Incidence of infection

The mean infection rate will be compared by trial arm using Poisson regression. The length of observation time per CHR will be adjusted. Results will be presented as the means, standard errors, adjusted IRR between trial arms, along with 95% CI and p-value.

Site of infection

Separate Poisson regression models will be fitted to compare the mean infection rate for each type of infection (UTI, RTI, skin, GI, unexplained fever). The length of observation time per CHR will be adjusted. Results will be presented as the adjusted IRR between trial arms, along with 95% CI and p-value.

Mean duration of infections

A comparison of the mean duration of infections will be carried by fitting a two-level linear regression model, with participants nested within care homes. The adjusted mean difference will be provided along with 95% CI and p-values.

Cumulative number of infection days

The cumulative number of days that participant experienced a suspected infection will be compared between trial arms using Poisson regression, with the length of observation time per participant included in the model. The means, standard errors, adjusted IRR between treatment arms will be provided alongside 95% CI and p-values.

Incidence of antibiotic-associated diarrhoea

The mean incidence rate of antibiotic-associated diarrhoea will be compared by trial arm using two-level Poisson regression. The length of observation time per CHR will be adjusted. Results will be presented as means and standard errors, the adjusted IRR between trial arms, along with 95% CI and p-value.

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Cumulative number of antibiotic-associated diarrhoea days

The cumulative number of antibiotic-associated diarrhoea days will be compared by fitting a two-level Poisson regression model. The means, standard errors, adjusted incidence rate ratio will be provided along with 95% CI and p-value.

Health utility (EQ-5D-5L)

Health utility will be compared between trial arms by fitting two two-level linear regression models to scores on the EQ5D scale at three months and final follow-up individually, with participants nested within care homes. Baseline reported EQ-5D-5L scale will be controlled for as a covariate. The adjusted mean difference will be provided alongside 95% CI and p-values. Proxy- and self-completed versions will be analysed separately.

Health status (EQ-VAS)

In a similar way, the EQ Visual Analogue scale (EQ VAS) will be compared between trial arms by fitting two two-level linear regression models to scores on the EQ VAS at three months and the final follow-up. Baseline reported EQ-VAS will be controlled for as a covariate. Proxy- and self-completed versions will be analysed separately. The adjusted mean difference will be provided alongside 95% CI and p-values.

Well-being (ICECAP-O)

In a similar way to the above, well-being will be compared between trial arms by fitting two two-level linear regression models to scores on the ICECAP-O scale at three months and the final follow-up. Proxy- and self-completed versions will be analysed separately.

Ever been hospitalised

The proportion of participants who have reported hospitalisation over the follow-up period will be estimated using a two-level logistic regression model. The adjusted odds ratio (OR) will be provided alongside 95% CI and p-values.

Cumulative number of hospital days

The cumulative number of hospital days will be compared between arms by fitting a two-level Poisson regression model. The means, standard errors, adjusted incidence rate ratio will be provided along with 95% CI and p-value.

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Death

The proportion of participants who die will be compared between arms using a two-level logistic regression model. The adjusted OR will be provided alongside 95% CI and p-values. A two-level frailty model will also be fitted to compare the time from randomisation to death between arms.

6.2.2 COVARIATE ADJUSTMENT

All analyses will adjust for gender of the participant.

6.2.3 ASSUMPTION CHECKING

Appropriate checking of the data and models used in the primary and secondary analyses will be carried out.

For count outcomes that will be modelled using Poisson regression, negative binomial regression will be used in the presence of over-dispersed count data.

6.2.4 ALTERNATIVE METHODS IF DISTRIBUTIONAL ASSUMPTIONS NOT MET

Negative binomial regression will be used in the presence of over-dispersed count data.

Appropriate transformations will be used to assumptions of linear regression are fulfilled for the EQ-5D-5L, EQ-VAS, and ICECAP-O mixed models. If transformation is not applicable (e.g. due to excess of particular values, such as values of 1 for the EQ-5D-5L utility score), a two-part Hurdle model will be fitted.

6.2.5 SENSITIVITY ANALYSES

Sensitivity analysis will be carried out by counting the number of hospital stays using a Poisson regression model. Primary analysis will be rerun by including antibiotics that prescribed as prophylaxis antibiotic in the CAAD outcome. Other sensitivity analyses were described within the analysis and missing data sections (see sections 6.2.1 and 6.3).

The impact on the primary analysis of ignoring periods of hospitalisation for which sufficient data are not available to calculate CAAD during these periods (i.e. assumption 1 in Appendix III) will be explored.

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6.2.6 SUBGROUP ANALYSES

Subgroup analyses will be performed to explore differential treatment effects on the primary outcome measure. Interaction terms will be fitted between trial arm and the following measures thought to be correlated with the primary outcomes. These include:

- Gender
- Baseline capacity to consent to the trial
- Clinical frailty scale at baseline (grouped as very fit to managing well / vulnerable to moderately frail / severely frail to terminally ill)

6.3 MISSING DATA

By including the time at risk as an offset variable in our primary analysis, this will automatically account for missing data.

It is likely that the majority of missing data will be due to drop-out, with reasons for dropout falling into two broad categories, withdrawal from the trial and death. An examination of the distribution and amount of missing between trial arms, by care home and reason will be tabulated.

A comparison will be made using baseline variables between those for which primary outcome and selected secondary outcome (depending on the volume of missing) data is and is not available. Appropriate modelling techniques will be used (e.g. likelihood-based methods or multiple imputation). Sensitivity analysis will be conducted using joint modelling approaches (e.g. selection and/or pattern mixture models) to explore departures from the missing at random assumption.

For additional sensitivity analyses, a comparison between participant death during an infection and death from other cause will be explored. An approach which is valid under a missing at random assumption will be applied to the deaths from any other cause and take a more extreme approach with those that died during an infection (e.g. assume that they would have remained on antibiotics for the remainder of the follow-up period).

Similarly, we will explore stratifying drop-out due to residents moving to another care facility into two groups: those who move for health-related reasons (e.g. from a residential home to a care home providing nursing care) and those who move for other reasons. A missing at random-valid modelling approach will be applied to the latter strata, with a more extreme approach taken for those dropping out due to moving from the care home for health-related reasons as this represents a decline in health, which may also be subject to unmeasured selection bias.

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6.4 ADDITIONAL ANALYSES

6.4.1 ADHERENCE-ADJUSTED ANALYSIS

Structural mean models will be fitted in order to obtain adherence-adjusted estimates of the primary and secondary analyses. Depending on the type of outcome, these models will be fitted using the generalised methods of moments framework (for generalised linear models) or two-stage least squares procedure (for linear models).

Two definitions of study product adherence will be used for this analysis:

- Initiation – as described in 4.2.2
- Percentage of days that participants took their study product as prescribed (with those not initiating set to zero)

The model will include gender as a stratification variable and use robust standard errors to account for clustering effects by care homes.

6.4.2 DAILY STUDY PRODUCT IMPLEMENTATION OVER TIME

A three-level logistic regression model (responses nested within participants within care homes) will be fitted to daily study product use (study product taken / study product not taken). Time will be included as a covariate, with non-linear patterns of adherence over time accounted for using splines, if necessary.

6.4.3 ASSESSMENT OF BLIND BEING BROKEN

At the final follow-up, care home staff will be asked what trial arm they think the participant is in (active, placebo, or no opinion). The agreement between response from care home staff and actual allocation will be assessed using James' blinding index. (13) Summary statistics will be described with respect to overall, trial arm and care home.

6.4.4 METHOD OF INGESTION

The method of ingestion (swallowed with water, sprinkled on food and sprinkled on drink) will be collected over the 12 months study period. Frequency of each method will be reported by overall and split by care home.

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6.4.5 ESTIMATING HEIGHT FROM ULNA LENGTH

An exploratory analysis will be carried out to assess the agreement between measured height and height predicted from ulna length. Participants who are unable to measure height or ulna length will be excluded from the analysis. Appropriate descriptive summaries and graphical illustrations of baseline demographic including age, gender, ethnicity and clinical frailty scale will be presented split by those who are included and excluded from the analysis to assess the missing pattern.

Participant's height will be measured in metres, and ulna length will be measured in centimetres. The predict height and ulna length will be converted using the Malnutrition Universal Screening Tool (MUST) table. Summary statistics (n, minimum, maximum, mean, SD) on the difference between these two measures and the measured and predicted values using the MUST table will be described with respect to gender and overall. The correlation between the different methods will be evaluated using intraclass correlation coefficient (ICC) to assess the agreement between these two methods. Bland–Altman analysis will be used to assess any systematic differences between the two methods, with 95% limits of agreement also calculated.

6.4.6 ESTIMATING BMI FROM MID UPPER ARM CIRCUMFERENCE

In a similar way, an exploratory analysis will be carried out to assess the agreement between measured body mass index (BMI) categories and BMI categories predicted from mid upper arm circumference (MUAC).

Participant's weight will be measured in kilograms. BMI will be calculated using formula: $\text{weight (kg)} / [\text{height (m)}]^2$. The predicted BMI categories and MUAC will be converted using the MUST tool. Summary statistics (n, sensitivity, specificity) on the measured and predicted categories will be described with respect to gender and overall. Agreement will be by calculating using weighted kappa statistics.

6.4.7 AGREEMENT BETWEEN METHODS USED TO MEASURE STUDY PRODUCT USE

We will conduct exploratory analysis comparing the agreement between study product use as assessed by weekly records (i.e. primary measure of study product use) and capsule counts. We will also investigate participant and care home predictors of disagreement. The model will account for the presence, direction, and extent to which there is disagreement.

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6.4.8 RELATIONSHIP BETWEEN NUTRITIONAL STATUS, FRAILTY, CHANGE IN CAPACITY STATUS, AND RISK OF DEATH

We will conduct exploratory analysis to investigate the association between the risk of death and changes in nutritional status, frailty, and capacity.

6.4.9 AGREEMENT BETWEEN SELF-REPORT AND PROXY EQ-5D AND ICECAP-O MEASURES

For participants with both self-report and proxy EQ-5D (utility and VAS) and ICECAP-O measures, we will assess the agreement between these two measures by calculating weighted kappa statistics, intraclass correlation coefficients, and Bland-Altman plots for categorical and continuous measures respectively.

6.5 HARMS

The number of serious adverse reaction (SAR) will be reported by their relationship as 'definitely' and 'probably' related to the study product. The number of participants experiencing SAE will be compared descriptively by trial arms. For each participant, only the maximum severity experienced SAE will be displayed. The number of occurrences of each SAE will also be presented for each trial arm. No formal statistical testing will be undertaken.

6.6 STATISTICAL SOFTWARE

Data will be cleaned and managed using IBM SPSS Statistics version 20 or later. Formal statistical analyses will be performed in Stata version 13 or later.

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7. REFERENCES

7.1 NON STANDARD STATISTICAL METHODS

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9. Vrijens B, De Geest S, Hughes DA, Przemyslaw K, Demonceau J, Ruppar T, et al. A new taxonomy for describing and defining adherence to medications. *British journal of clinical pharmacology*. 2012;73(5):691-705.
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11. Moher D, Hopewell S, Schulz KF, Montori V, Gøtzsche PC, Devereaux P, et al. CONSORT 2010 explanation and elaboration: updated guidelines for reporting parallel group randomised trials. *Journal of clinical epidemiology*. 2010;63(8):e1-e37.
12. Akaike, H., 1974. A new look at the statistical model identification. *IEEE transactions on automatic control*, 19(6), pp.716-723.
13. Bang, H., Ni, L. and Davis, C.E., 2004. Assessment of blinding in clinical trials. *Controlled clinical trials*, 25(2), pp.143-156.

7.2 DATA MANAGEMENT PLAN

The PRINCESS data management plan is version 1.0 located here

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S:\PCAPH\PCAPH\SEWTU Studies\PRINCESS\15. Data management & IS\Data management plan\Current version

7.3 TRIAL MASTER FILE AND STATISTICAL MASTER FILE

The trial master file is located in the following location:

S:\PCAPH\PCAPH\SEWTU Studies\PRINCESS

7.4 OTHER SOPs OR GUIDANCE DOCUMENTS

International conference on harmonisation, editor. Harmonisation tripartite guidelines E9 – Statistical Principles for Clinical Trials, Version 4: [<http://www.ich.org/LOB/media/MEDIA485.pdf>]5th Feb 1998.

International conference on harmonisation, editor. Harmonisation tripartite guidelines E6(1) – Guidelines for good clinical practice, Version 4: [<http://www.ich.org/LOB/media/MEDIA482.pdf>]10th June 1996.

International conference on harmonisation, editor. Harmonisation tripartite guidelines E3 – Structure and content of clinical study reports: [<http://www.ich.org/LOB/media/MEDIA479.pdf>]30th Nov 1995.

Council DB, Hall T, House W, Lane W. Data Protection Act 1998. 2011.

SOP/008/1 – Design and implementation of a randomisation strategy template

SOP/008/2 – Statistical Analysis Plan

SOP/008/3 – Sample Size Calculations

SOP/008/4 – Statistical Reporting

SAP/ISAP DEVIATION LOG

Document number:		Document version:	
Reason for deviation:			

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8. APPENDICES

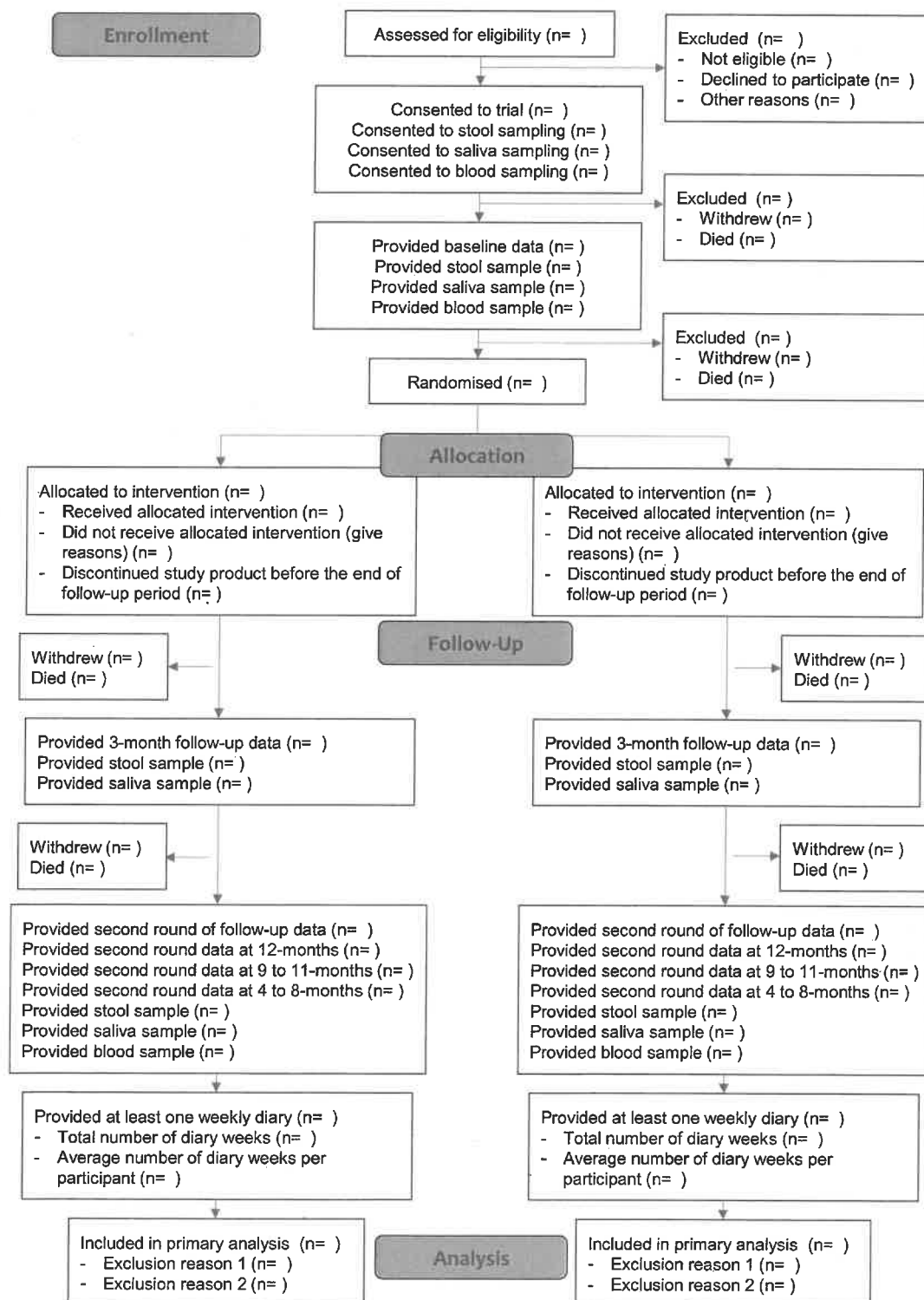
Appendix I: Schedule of study procedures

Assessment	Baseline	Once Daily (for 12 months)	Regular intervals (for 12 months)	Monthly (for 12 months)	If vaccinated against flu, on day of vaccination or as close as possible	If vaccinated against flu, 4 weeks (28 days) afterwards	3 month*	Second follow- up*
Assessment of eligibility	X							
Informed consent or consentee declaration"	X							
Demographic data	X							
Brief clinical history	X							
Clinical Frailty Score	X							
CAAD for infection			X				X	X (from MAR and hospital record, if applicable)
Infection details			X				X	X
Oral rinse or saliva culture" for oral candida	X						X	X
Blood sample" (FBC, Vitamin D)	X							
Blood sample" for serum cytokines (ex-vivo response) (n=100)	X							X plus FBC blood sample
Blood sample" (influenza vaccine efficacy)					X	X		
Stool sample" (C.diff, antibiotic sensitivity of gram -veenterobacteria and VRE, gut microbiota diversity or phylogeny)	X						X	X
EQ-5D" (participant and/or proxy completion)	X						X	X
ICECAP-O" (participant and/or proxy completion)	X						X	X
NHS service use (Care Home note review)			X					X
NHS service use (Hospital discharge summaries)								(X)
Mortality								X
Randomisation	X							
Study product pack allocation				X				
Study Product administration		X						
Adherence to Study Product (MAR sheet)		X (MAR sheet completion as often as practicable)	X (transcri be to CRF)					

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Appendix II: Example CONSORT flow diagram

CONSORT 2010 Flow Diagram



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Appendix III: Dummy tables

Variable		Intervention (n=)		Control (n=)		Total (n=)	
		n	%	n	%	n	%
Age at baseline	Range						
	Mean (SD)/Median (IQR)						
Gender	Male						
	Female						
Ethnic group	Appropriate categorisation will be presented						
Height	Mean (SD)						
weight	Mean (SD)						
Ulna length (Left)	Mean (SD)						
Ulna length (Right)	Mean (SD)						
Mid upper arm circumference (Left)	Mean (SD)						
Mid upper arm circumference (Right)	Mean (SD)						
Clinical frailty scale	Mean (SD)/Median (IQR)						
Number of antibiotics prescribed in the last four weeks	Appropriate categorisation will be presented						
Used a proton pump inhibitor in the last four weeks	Yes						
	No						
Used a laxative in the last four weeks	Yes						
	No						
Used a vitamin D supplement in the last four weeks	Yes						
	No						
EQ-5D (5L)-proxy version	Mean (SD) / Median (IQR)						
EQ-5D (5L) – self-completed version	Mean (SD) / Median (IQR)						
ICECAP-O – proxy version	Mean (SD) / Median (IQR)						
ICECAP-O – self-completed version	Mean (SD) / Median (IQR)						

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Appendix IV: Assumptions about antibiotic administration while in hospital

The primary objective of the PRINCESS study is to evaluate the effect of a daily dose of oral probiotics on cumulative antibiotic administration days for all-cause infections during the 12-months post-randomisation. Any time a participant is prescribed an antibiotic for an acute infection, the number of days that the antibiotic is administered will be recorded. CAAD will be a summation of these days for the duration of the follow-up period. Hospitalisations are an expected event in our study population, and antibiotic administration is likely to be high while they are in hospital. Concern has been expressed about whether or not we will be able to discern the number of days that participants were administered antibiotics for an acute infection during their hospital stay. We have developed a hierarchy of assumptions that we will use when assessing CAAD during periods of hospital stays. See Table below for list of assumptions.

Table: Hierarchy of assumptions made from hospital discharge summaries

Assumption*	Description	Impact on cumulative antibiotic administration days outcome
1	Discharge summary letter provides detail of antibiotics being administered, clinical indication, and number of administration days	Use information on discharge summary
2	Discharge summary letter provides detail of antibiotics being administered and clinical indication (but not number of days)	If clinical indication is an acute infection, assume participant was administered antibiotics for 14 days or length of hospital stay (whichever is smaller)
3	Discharge summary letter provides detail of antibiotics being administered (but not clinical indication or number of days)	If they were admitted for a suspected acute infection (or symptoms associated with suspected acute infection [†]), assume participant was administered antibiotics for 14 days or length of hospital stay (whichever is smaller)
4	Discharge summary letter provides no detail of antibiotics being administered	If they were being administered antibiotics prior to admission and is discharged with an antibiotic prescription, assume

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Assumption*	Description	Impact on cumulative antibiotic administration days outcome
		participant was administered antibiotics for 14 days or length of hospital stay (whichever is smaller)

*Where 1 is the most and 4 is the least plausible assumption. †For example, high fever, dehydration.

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Appendix V: Assumptions about infections while in hospital

Similar to appendix III, as a key secondary outcome, we are also interested in capturing infections that occur during periods of hospitalisation.

In the Table below we list our hierarchy of assumptions we will make regarding infections during periods of hospitalisation.

Table: Hierarchy of assumptions made from hospital discharge summaries

Assumption*	Description	Impact on incidence of infections
1	Discharge summary or weekly diary provides information for a systemic infection while in hospital	Counts as infection
2	Discharge summary letter provides detail of antibiotics being administered for a systemic infection while in hospital (i.e. no indication of prophylaxis)	Counts as an infection – in separate treatments / episodes are indicated, count each episode
3	Discharge summary letter provides no detail of antibiotics being administered	If discharged with antibiotic prescription (not indicated as prophylaxis), then count an episode of infection occurring during period of hospitalisation

*Where 1 is the most and 3 is the least plausible assumption.

